The CTC Algorithm

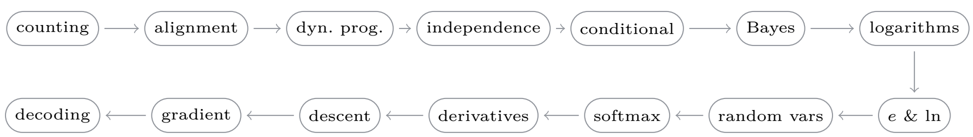
A study guide, stitched from the manim-concepts road

manim-concepts

Contents

How to read this guide

The objective is full comprehension of the CTC algorithm — the alignment machinery, the loss, the gradient, the training dynamics, and decoding. The road runs through fourteen stations; each is a chapter, and between chapters a short transition marks where you stand — the roadmap returns there with your progress filled in:



The order is the dependency order: the CTC chapters pose the problem early, the middle chapters build the blocks it needs, and the closing chapters spend them. Practice problems close every chapter; worked solutions live in the separate solutions manual, keyed by the same problem numbers. Worked numbers are spliced from the repo's verification anchors — for instance, the guide's central worked probability $P(\mathrm{AB} \mid X) = 0.4877$ — never retyped.

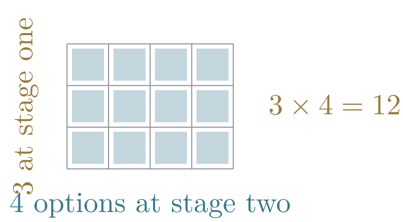
Chapter 1

Counting rules

Every question this book asks eventually reduces to *how many ways can this happen, and how heavy is each way?* The second half of that question is probability's business, and it comes later. This chapter is the first half alone: four counting rules, each built by counting real objects before any formula appears — and a closing observation that the four are secretly one rule wearing different amounts of correction.

1.1 Independent stages multiply

Suppose a choice happens in stages: first one of 3 options, then — whatever was chosen first — one of 4 options. Drawn as a tree, the first choice opens 3 branches and each branch opens 4 more: 12 leaves. Drawn better, the same count is a *grid*: stages become axes, and the total is literally the area of a 3×4 rectangle.



The grid earns the rule — the classical rule of product [50] — its condition: the stages must be *independent* in the counting sense — the *number* of options at stage two cannot depend on what stage one chose (the options themselves may differ; their count may not). When that holds,

$$N \;=\; n_1\, n_2 \cdots n_k.$$

The formula is the last thing on screen in the video, and the last thing in this section: first the leaves were counted, then the rectangle, and only then the product that summarises what just happened.

1.2 Order matters: the shrinking pool

Fill r slots from a pool of n distinct items, where filling a slot *removes* the item. Three podium places from five runners: the first slot has 5 choices, the second 4, the third 3 — a product again, but of a pool that shrinks: $5 \times 4 \times 3 = 60$. The factorial form is bookkeeping for exactly this: write 5! and cancel the tail you never used,

$$P_r^n = \frac{n!}{(n-r)!}, \qquad P_3^5 = \frac{5!}{2!} = 60.$$

Hold on to the shrinking pool itself — not just the formula. It is the picture of *sampling without replacement*, and several sections from now it will be the exact thing that breaks independence.

1.3 Order does not matter: divide out the over-count

Choose 3 letters from 5 where the *set* is the answer — $\{A, C, E\}$ chosen in any order is the same choice. The shrinking pool counted 60 ordered selections, but every set of 3 was counted once per ordering: $3! = 6$ times. The correction is a division [11],

$$C_r^n = \binom{n}{r} = \frac{n!}{r!\,(n-r)!}, \qquad \binom{5}{3} = \frac{60}{6} = 10.$$

This is the chapter's load-bearing move — *count with too much order, then divide out the order that changes nothing* — and it is the move the rest of the road keeps reusing.

1.4 Partitions: the same move, blockwise

Split 6 distinct items into labelled groups of sizes 3, 2, 1. Order all 6 (6! ways), chop the ordering into blocks — and notice that reordering *within* a block changes nothing. Divide out each block's internal orderings:

$$N = \frac{n!}{n_1!\,n_2!\cdots n_k!}, \qquad \frac{6!}{3!\,2!\,1!} = 60.$$

A combination is the special case with two blocks (chosen, not-chosen).

1.5 Which rule does a problem want?

The closing scene of the series maps problem shapes to rules, then closes on an observation worded with care: *multiply the choices — divide only when orderings mean the same outcome.* Three of the four rules are the product rule with unwanted orderings divided out afterwards; the product rule itself divides nothing. Ask two questions of any counting problem — *do the stages' option-counts depend on earlier choices?* (if yes, the pool is shrinking) and *does swapping two items change the answer?* (if no, divide the orderings out) — and the right rule falls out.

Where this goes. The grid returns weighted (areas instead of counts) when probability arrives; the shrinking pool returns as the thing that breaks independence; the binomial coefficient returns as a count of *sorted columns*; and the very next chapter opens a search space of size $3^4 = 81$ raw sequences — the product rule sizing a problem that will take half this book to finish solving.

Practice problems

Problem 1.1. A per-frame labeller writes one of 3 symbols at each of 4 frames, independently. How many distinct label sequences can it write?
Hint: One stage per frame; the option count never changes.

Problem 1.2. Eight runners; how many distinct podiums (gold, silver, bronze) are possible?
Hint: Slots that empty a shrinking pool — and the medals make order matter.

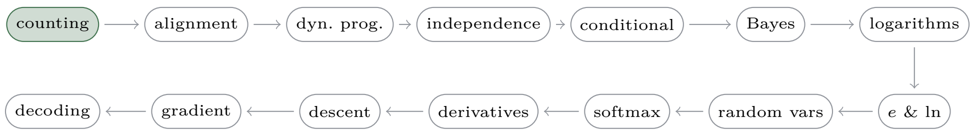
Problem 1.3. From the same eight runners, how many three-person relay *teams* (no roles) can be formed? Check your answer against the previous problem using the overcount argument.
Hint: Every team was counted once per internal ordering.

Problem 1.4. How many distinct arrangements does the word SEEDED have?
Hint: Six letters, but reordering equal letters changes nothing — divide blockwise.

Problem 1.5. [stretch] A walker on a grid moves from the bottom-left corner to the top-right corner of a 2-row-by-4-column lattice of steps, moving only right or up — 4 rights and 2 ups in some order. How many routes are there?

Hint: A route is a word in $\{R, U\}$ with repeated letters.

The road so far



You own two moves now: *independent stages multiply*, and *count with too much order, then divide the surplus out.* The last problem left you holding a number — 15 routes through a small lattice — that seemed to be about a walker.

The next chapter poses the problem this whole guide exists to solve, and your two moves are the first tools it reaches for: the product rule sizes a search space of $3^4 = 81$ raw sequences in one line, and the divide-out instinct is what makes the constrained count $\binom{T+U}{T-U}$ readable. The walker's lattice is about to reappear as a grid of *states against frames* — same fifteen, different costume — and the question will stop being "how many?" and become "how heavy is each one?". That second question is a debt the chapter takes on openly; most of this book is the repayment.

Chapter 2

The alignment problem, and CTC's answer

Here is the problem the whole road exists to solve. A speech clip is T audio frames; its transcript is U characters, with $U < T$ and no marking of which frames belong to which character. The transcript says *what* was said, never *when*. Any training signal that demands per-frame labels demands data nobody has.

Connectionist Temporal Classification (CTC) [17] is the standard answer, and this chapter builds its machinery — using, on credit, one notion the road has not yet earned: each frame t carries a score $y_t(k)$ for each symbol k, which we will treat as "the model's per-frame probability". What those scores are and why they behave like probabilities is exactly what the middle chapters of this guide construct; here they are simply weights the model hands us. (The video series carries the same declared debt in its Scope.)

2.1 The blank token, and the collapse map

Let the model emit one symbol per frame from the alphabet plus one invented symbol: the *blank* ε, "nothing new to emit". A raw emission sequence — a *path* — becomes a transcript by the collapse map $\mathcal{B}$: **merge** adjacent repeats, **then drop** blanks. The order is load-bearing: merging first lets a blank protect a genuine double letter (HE LεL O survives as HELLO), while dropping first would weld the two L's into one. Without the blank, a per-frame emitter with repeat-merging can never write a double letter at all.

2.2 A transcript's probability is a sum over paths

Fix the worked example the whole road shares: alphabet $\{A, B, \varepsilon\}$, transcript Y = AB, and $T = 4$ frames. Many paths collapse to AB — AAεB, εABB, ABBε, … — and the data cannot prefer one of them, so the model must not be forced to. CTC's defining move: score the transcript by *every* path that spells it,

$$P(Y \mid X) \;=\; \sum_{\pi \in \mathcal{B}^{-1}(Y)} \prod_{t=1}^{T} y_t(\pi_t).$$

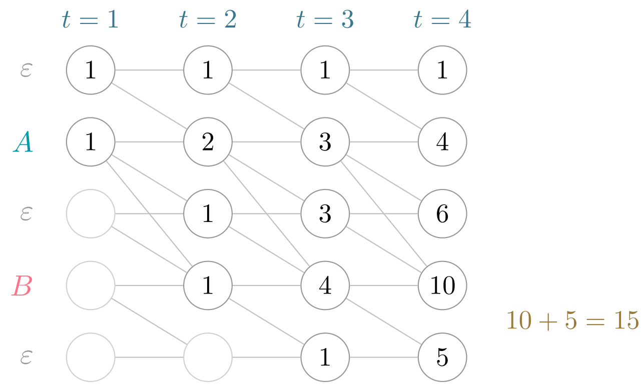
Each path's weight is a product of per-frame scores (the frames are treated as independent given the input — a real assumption, examined at the chapter's end), and the transcript's probability is the sum of those products. Not the best path: the sum. Greedy decoding — take each frame's favourite symbol and collapse — can return a transcript that no single path dominates but the sum ranks first; the sum is the object CTC trains on.

2.3 How big is the sum?

The product rule from the last chapter sizes the raw space instantly: 3 symbols over 4 frames is $3^4 = 81$ paths. Of those, exactly 15 collapse to AB — countable by hand, and counted on screen in the video. In general the repeat-free count is $\binom{T+U}{T-U}$, and at a realistic $T = 100$, $U = 50$ that is roughly 2×10^{40} — enumeration is dead on arrival. A sum this size must be *reorganised*, not listed.

2.4 The forward trellis

Wrap the transcript in blanks — AB $\rightarrow (\varepsilon, A, \varepsilon, B, \varepsilon)$, giving $2U + 1 = 5$ states — and lay the states against the frames as a grid. A path is a walk down the grid's legal edges: *stay* in a state, *advance* to the next, or *skip* over a blank — legal only between different letters, because skipping the blank between equal letters would merge them.



Paths sharing a prefix share their count — that is the reorganisation [22]. Let $\alpha_t(s)$ carry the weight of every partial path reaching state s at frame t; each column needs only the column before it,

$$\alpha_t(s) \;=\; \left(\alpha_{t-1}(s) + \alpha_{t-1}(s{-}1) + \alpha_{t-1}(s{-}2)\right) y_t(z'_s),$$

with the $s{-}2$ skip term present only where the grid allows it, and the transcript's total read off the two final states. At unit weights the recurrence is pure counting and lands on the same 15 as the hand count — two routes, one answer. Swap the units for the per-frame scores and the same grid computes $P(Y \mid X)$ exactly: an astronomically wide sum, priced on twenty nodes.

2.5 Where CTC applies — and where it cannot

Three assumptions are the trellis's shape: alignment is *monotonic* (audio does not reorder), the output is *no longer* than the input, and frames are *independent given the input.* Monotonic transduction — speech, handwriting, OCR — fits. Translation reorders, so it does not; that failure belongs to attention models. And the independence assumption leaves language on the table, which is why real systems add an external language model. One reading habit, now, before it can calcify: a trained CTC model's output spikes are *not* segment boundaries — why they look the way they do is this guide's final chapter.

Where this goes. The per-frame scores are debt: the middle chapters build probability from the ground up until softmax and the likelihood lens make y_t honest. The trellis returns twice more — once named for what it is (dynamic programming), and once swept backward, when the gradient chapter differentiates this very sum.

Practice problems

Problem 2.1. Apply the collapse map: what are $\mathcal{B}(\text{AA}\varepsilon\text{B})$, $\mathcal{B}(\varepsilon\text{ABB})$, and $\mathcal{B}(\text{ABBA})$?

Hint: Merge adjacent repeats first, then drop blanks.

Problem 2.2. Show by example that the collapse order matters: find a path that yields a double letter under merge-then-drop but loses it under drop-then-merge.

Hint: Put a blank between two equal letters.

Problem 2.3. Over $T = 3$ frames with alphabet $\{A, \varepsilon\}$, how many raw paths are there, and how many collapse to the transcript "A"?

Hint: The raw space is the product rule; for the collapse count, either enumerate the eight paths or run the unit-weight trellis on $(\varepsilon, A, \varepsilon)$.

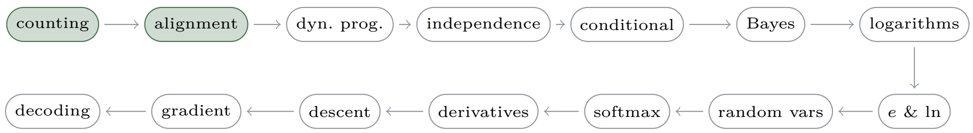
Problem 2.4. Using the repeat-free count $\binom{T+U}{T-U}$, how many paths spell a 2-letter transcript with no repeated letter at $T = 5$?

Hint: Substitute and compute; the answer also appears in the gradient chapter's dominance table.

Problem 2.5. [stretch] The chapter claims greedy decoding can be wrong. Over the two-symbol alphabet $\{A, \varepsilon\}$ at $T = 2$, construct per-frame scores where the frame-wise argmax path collapses to a transcript that is NOT the highest-probability transcript.

Hint: Give the blank a narrow win at every frame, and let the letter's many paths pool their weight against the blank's single path.

The road so far



You watched a sum too wide to list get priced on twenty nodes, and you watched the chapter borrow: those per-frame scores are still unearned. Before the road starts repaying, one short chapter stops to name the trick that just happened — because the trellis was not a lucky arrangement of circles. It was an instance of a general move, one you will use twice more before the end, and naming it now means recognising it instantly when it returns wearing weights.

Chapter 3

Dynamic programming

The last chapter ended on a sleight of hand worth catching red-handed. A sum over more paths than anyone could list was priced exactly on a grid of twenty circles, and the move was introduced in a single clause — *paths sharing a prefix share their count* — as if that were a small bookkeeping remark. It is not small. It is one sighting of a general method with its own name, its own conditions, and a habit of turning impossible counts into short columns of additions. This chapter takes the trellis back out and names what it did: *dynamic programming* — solve every small version of the problem once, store the answers, and let the big version assemble itself from the store.

3.1 The same question, asked over and over

Start away from CTC, with the smallest example that misbehaves. The Fibonacci numbers obey $F_n = F_{n-1} + F_{n-2}$ with $F_0 = 0$, $F_1 = 1$. Computed exactly as written — each F_n asking for its two predecessors, each of those asking again — the questions repeat: F_{10} asks for F_8 twice (once itself, once through F_9), for F_7 three times, for F_6 five times; the repetition compounds at the same rate as the sequence itself. Answering F_{10} this way takes 177 function calls to settle 11 distinct questions.

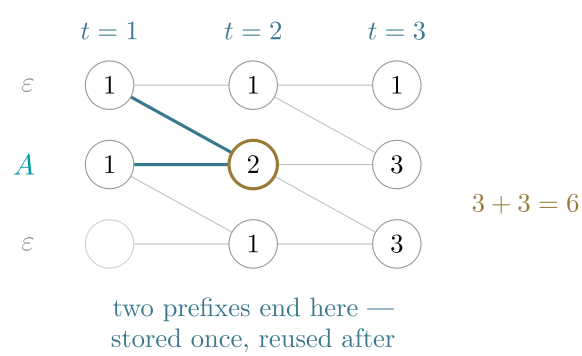
The cure is almost insultingly simple: *write the answers down.* Compute $F_2, F_3, \ldots$ in order, each from two stored numbers — eleven entries, one addition each. The exponential call tree collapses because two things were true at once. First, the subproblems *overlap*: the same question recurs, so a stored answer keeps getting reused. Second, the stored answer is *all the future needs*: whoever asks for F_8 cares only about its value, never about how the additions inside it were arranged. When both hold, the tree of recomputation flattens into a table.

3.2 The lattice, in its third costume

The counting chapter closed on a walker crossing a lattice — 4 rights and 2 ups, $\binom{6}{2} = 15$ routes, counted there by choosing which steps go up. Count it again, the dynamic-programming way: label every node with *the number of routes that reach it*. A node can only be entered from its left or lower neighbour, so its label is the sum of two earlier labels — Pascal's addition, marching across the grid. The corner lands on the same 15, and no route was ever written out. Each label is a stored subproblem: all routes agreeing on where they currently stand are carried as one number from there on. The exponential object (the list of routes) was reorganised into a small table of counts — *shared prefixes, counted once.*

3.3 The trellis, named

Now reread the last chapter with these words available. The mini trellis from its third problem — transcript A over $T = 3$ frames, states $(\varepsilon, A, \varepsilon)$ — enumerated $2^3 = 8$ raw paths and tested each against the collapse map. The recurrence answered the same question by filling nine cells:



The highlighted cell is the whole method in one circle. Two partial paths — εA and AA — arrive at state A after two frames. From that cell onward their futures are identical: what a path may legally do next depends only on *where it stands*, not on how it got there. So the two are stored as one number, and every continuation is counted once instead of twice. That is the memo table again: $\alpha_t(s)$ *is* the stored answer to "how much partial weight reaches state s by frame t?", and the recurrence builds each column from the stored one before it.

The same accounting, scaled: the AB trellis handled 81 raw paths, 15 of them accepted, on twenty cells. At $T = 100$, $U = 50$, the path count is roughly 2×10^{40} — and the trellis is 100×101 cells, about ten thousand entries of three additions and a multiplication each. The exponential object was never listed; it was reorganised. Both of dynamic programming's conditions did real work here: the subproblems overlap (astronomically many paths route through each cell), and the cell is all the future needs (the collapse map's legality — stay, advance, skip — reads only the current state). Had the continuation rules depended on more of the past, each cell would have had to remember more, and the grid would have grown states until they did.

3.4 Recognising a dynamic program in the wild

The signature has two parts, and both must be present. There is a brute-force count, sum, or best-choice over exponentially many arrangements; and there is a small *state* — the walker's node, the trellis cell, the index n — such that arrangements agreeing on the state have interchangeable futures. Edit distance between two strings (the state: a prefix of each), shortest routes through a road network (the state: the junction you stand at), and the forward algorithm this road met last chapter — the same reorganisation runs hidden-state models generally [37] and is the standard route through CTC [22]. The boundary is just as useful to know: if subproblems never repeat, a memo stores answers nobody asks for again — splitting a list into disjoint halves is divide-and-conquer, not dynamic programming — and if no small state summarises the past, the table itself grows exponential and the method returns nothing.

Where this goes. The trellis is not finished with this road. When the gradient chapter differentiates the alignment sum, the same grid will be swept *backward* — a second dynamic program, over suffixes — and the two sweeps together price every path through every cell. Later, the logarithms chapter will find the trellis's long products starving floating point, and the recurrence will move into log-space to survive. And this chapter itself is a seed: the video road performed dynamic programming without ever naming it, and a future series owes the concept its own screen time — these pages are its first draft.

Practice problems

Problem 3.1. Computing F_{10} by the bare recurrence $F_n = F_{n-1} + F_{n-2}$ (with $F_0 = 0$, $F_1 = 1$ answered directly), how many function calls are made in total? How many *distinct* questions were asked?

Hint: Count calls with their own recurrence: $c_n = 1 + c_{n-1} + c_{n-2}$, with $c_0 = c_1 = 1$.

Problem 3.2. Label every node of the 4-right, 2-up lattice with the number of routes reaching it, using only the addition rule. Confirm the far corner agrees with the counting chapter's $\binom{6}{2}$.

Hint: Start with 1 at the origin; every other node is the sum of its left and lower neighbours (a missing neighbour contributes nothing).

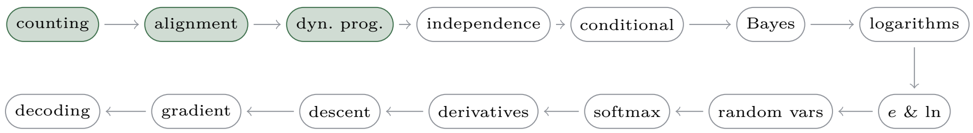
Problem 3.3. Extend the mini trellis one more frame: for transcript A over $T = 4$ frames, compute the fourth column from $(1, 3, 3)$ and read off the accepted count. Then confirm the count by a direct argument about which raw paths collapse to A.

Hint: Stay and advance only — no skip exists on $(\varepsilon, A, \varepsilon)$. For the direct argument: a path collapsing to A shows one contiguous block of A's.

Problem 3.4. [stretch] At $T = 100$ frames and $U = 50$ letters, compare the work the two routes perform: how many cells does the trellis fill, and how does that stand against the raw repeat-free path count? What would happen to the table if a path's legal continuations depended on its last *two* states instead of one?

Hint: The wrapped transcript has $2U + 1$ states. For the second part: what must a cell now remember?

The road so far



The move has a name now, and the grid is honest about what it computes. What it is not honest about yet is what it multiplies: a path's weight is a product of per-frame scores, taken on faith. Whether the world permits that multiplication — when a joint chance really is a product of separate chances — is not bookkeeping; it is the first real claim about probability this road makes, and the next chapter builds the square that decides it.

Chapter 4

Independence

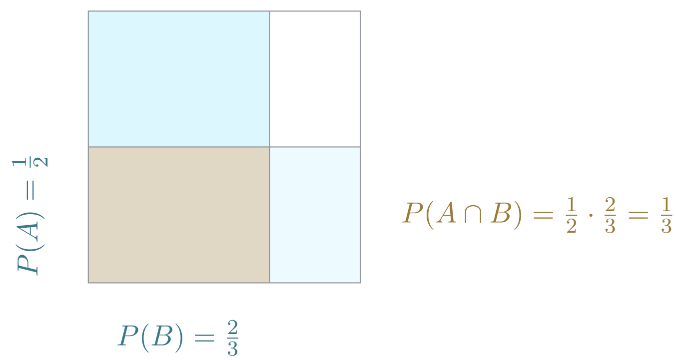
The counting chapter multiplied option counts and owed nothing for it. The alignment chapter multiplied per-frame scores into a path's weight — and flagged the move as an assumption it could not yet defend. This chapter is where multiplying becomes lawful for probabilities: what the product of two probabilities actually asserts, how to check it, the two confusions it attracts, and what assuming it costs.

4.1 Probability is area

Draw the sample space as a 1×1 square: an *event* is a region, and its probability is its share of the area [6]. A fair die is six equal cells; the event "even" covers three of them, and $3/6$ — count over total — is the same number as the region's area. Nothing new was computed: proportion is what counting was already doing, divided through by the size of the whole. The math of probability is the math of proportions, and every argument in this chapter is an argument about regions of this square.

4.2 The product rule

Run two experiments — a coin and a die, two dice, two frames. The counting chapter's grid returns with its cells reweighted from counts to areas: an event about the *first* experiment selects rows, an event about the *second* selects columns, and their intersection is a rectangle. Rectangle area is width times height — that is the whole proof. On the unit square, both cuts run straight across, all the way [53].



When the picture looks like this — each event a straight band, the overlap a rectangle — the joint probability factors, and that factoring is the *definition* of independence [57]:

$$A \perp B \quad \Longleftrightarrow \quad P(A \cap B) \;=\; P(A)\, P(B).$$

As in the counting chapter, the formula arrives last: first the grid, then the rectangle, then the equation that summarises what the picture already said. The definition is the product form itself — not a statement about causes, mechanisms, or time.

4.3 One die, two events

Independence does not need two experiments. On a single fair die, take $A = \{\text{even}\}$ and $B = \{1, 2, 3, 4\}$. Then $A \cap B = \{2, 4\}$, and

$$P(A \cap B) = \tfrac{2}{6} = \tfrac{1}{3} = \tfrac{1}{2} \cdot \tfrac{2}{3} = P(A)\, P(B)$$

— independent, though both events are about the same roll. Now slide B's boundary a single pip, to $\{1, 2, 3\}$: the intersection shrinks to $\{2\}$, and $\frac{1}{6} \neq \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$. Dependent. One pip decided it. And the same two events that factored on the fair die fail to factor on a biased one: independence is a property of the *measure* — of how the weight is spread — not of the events alone [35]. "Separate mechanisms" is neither necessary for independence nor sufficient; the arithmetic is the only test.

4.4 Mutually exclusive is not independent

The most-tested confusion in probability: surely events that cannot co-occur are "unrelated"? The product test says the opposite, as loudly as it can. If

A and B are disjoint with positive probability, then

$$P(A \cap B) \;=\; 0 \;\neq\; P(A)\,P(B) \;>\; 0.$$

Disjoint regions share no area, so the left side is zero while the right side is not — mutually exclusive events are *maximally* dependent: seeing one tells you the other did not happen. On the die, $\{\text{even}\}$ and $\{1, 3, 5\}$ fail the test at 0 versus $\frac{1}{4}$. (One honest exception: an event of probability zero is trivially independent of everything.) This is also why Venn-style pictures mislead here — disjoint circles *look* unrelated. The square tells the truth the circles hide: dependence is the cut through the band stepping; independence is the knife-edge case where it runs straight.

4.5 Chains of trials

One flip cuts the square in half. A second flip cuts each half in half again; after four flips the square holds sixteen cells, and the particular sequence HHTH is one cell of area $1/16$ — each independent trial multiplied one more factor on. In general, for independent events $A_1, \ldots, A_n$,

$$P(A_1 \cap \cdots \cap A_n) \;=\; \prod_{i=1}^{n} P(A_i),$$

a box with side lengths $p_1, \ldots, p_n$. Two honest captions come with the chain. First, it needs *mutual* independence, which is strictly more than every pair factoring: with two fair coins, take $A = \{\text{first is H}\}$, $B = \{\text{second is H}\}$, $C = \{\text{exactly one head}\}$ — all three pairs factor at $\frac{1}{4}$ each, yet the triple intersection is empty, $0 \neq \frac{1}{8}$ (Bernstein's construction [60]). Second, asserting the product over a real sequence of trials is a *modeling choice* — the product measure [43] — not an observation. This is precisely the purchase the alignment chapter made on credit when it priced a path as a product of per-frame scores: one factor per frame is this formula, asserted about a model.

4.6 When to use it

The chapter's exit is a field guide. *Replacement installs independence*: draw a card, put it back, shuffle — the second draw faces the same deck, and two aces cost $1/169$. *Depletion breaks it*: keep the first card out and the same two aces cost $1/221$ — the counting chapter's shrinking pool, returned exactly as promised, as the thing that breaks independence. *Common causes break it too*, with no causal link between the events themselves: two symptoms of one disease co-occur far more often than their product predicts. The alignment chapter already met this in the wild — within a word, the frames of speech share the word as a common cause, which is why real systems bolt an external language model onto a per-frame emitter at decode time. And the *gambler's fallacy* is independence misread in reverse: a run of heads owes the future nothing; the long-run proportion settles by *swamping* — ever more trials diluting any early excess — not by compensation [40]. Independence is a property of the model's measure, and assuming it is a purchase. Know what you paid.

Where this goes. The next chapter buys back this chapter's shortcuts: the stepped cut gets a name — conditional probability — the aces' 1/221 finally gets its license as a product of a probability and a *conditional* probability, and the assumption the alignment chapter actually needs is stated in full: frames independent *given the input*. Further down the road, the sixteen-cell square built here is stamped and sorted into a distribution, and the per-frame scores become honest probabilities.

Practice problems

Problem 4.1. Roll two fair dice. Let $A = \{\text{first die shows 5 or 6}\}$ and $B = \{\text{second die shows 3 or less}\}$. Compute $P(A \cap B)$ and check it against the product rule.

Hint: Draw the 6×6 grid; A selects rows, B selects columns.

Problem 4.2. On one fair die, test $A = \{\text{even}\}$ for independence against $B = \{1, 2, 3, 4\}$ and against $B' = \{1, 2, 3\}$.

Hint: Three probabilities each time; the definition is the only test.

Problem 4.3. Show that $A = \{\text{even}\}$ and $B = \{1, 3, 5\}$ on a fair die are not independent, and say how much information seeing B carries about A.

Hint: What is the area of a disjoint overlap?

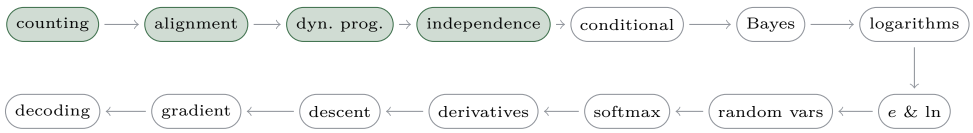
Problem 4.4. Flip two fair coins. Let $A = \{\text{first is H}\}$, $B = \{\text{second is H}\}$, $C = \{\text{exactly one head}\}$. Verify that every pair is independent, and that the three events are not mutually independent.

Hint: Four outcomes, each of probability $\frac{1}{4}$; check the three pair products, then the triple.

Problem 4.5. [stretch] Draw two cards from a shuffled deck. Compute $P(\text{both aces})$ with replacement and without, and — without replacement — $P(\text{second card is an ace})$ before anyone looks at the first.

Hint: Replacement makes the draws independent trials; depletion is the shrinking pool. For the last part, enumerate — or argue by symmetry.

The road so far



Independence gave the product rule a picture: straight cuts on the unit square, area times area. It also showed you the picture breaking — the shrinking pool, the aces priced at anchor002.aces.depleted with no license shown. The next chapter issues the license: the stepped cut gets a name, the price gets a factorisation, and the alignment chapter's exact assumption — independent emphgiven the input — finally gets stated in full.

Chapter 5

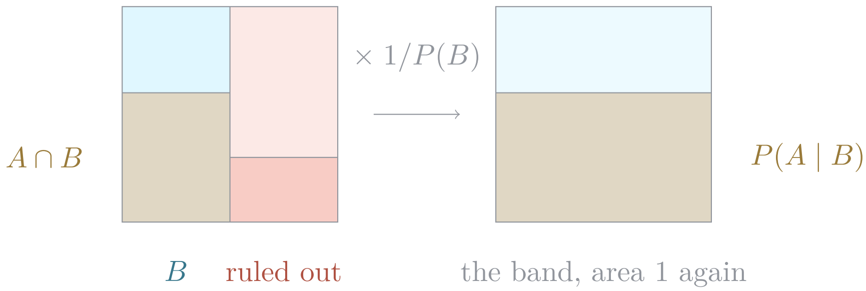
Conditional probability

Every probability so far was measured against the whole square. But most questions arrive mid-story: *given that* the first coin came up heads, *given that* the test came back positive, *given that* the first card was an ace. This chapter builds the tool for "given that" — and in doing so pays two debts at once: the independence chapter's stepped cut finally gets its name, and the aces' price tag finally gets its license.

5.1 Restriction, then renormalization

Flip three fair coins. Three heads is one outcome of eight: $\frac{1}{8}$. Now learn that the first coin landed heads. Half the outcomes are ruled out; among the four that remain, three heads is one: $\frac{1}{4}$. No formula was needed — throw away what the information rules out, recount inside what is left.

On the square the same move is geometric. The event B is a band; dim everything outside it — ruled out, discarded — and stretch the band back to a full unit square. The stretch multiplies every area inside the band by exactly $1/P(B)$, so every *within-band* proportion is preserved: the slice is a genuine probability space in its own right. "Conditional probabilities are probabilities" [9].



The formula is the picture's bookkeeping, which is why it comes last:

$$P(A \mid B) \;=\; \frac{P(A \cap B)}{P(B)}, \qquad \text{defined only for } P(B) > 0.$$

Restrict to B (the numerator keeps only the part of A inside the band), renormalize (divide by the band's area). Nothing else happened.

5.2 The step was conditional probability all along

The independence chapter drew dependence as a cut that *steps* at the band's edge and independence as the cut running straight — and left the step unnamed. Here is its name: the height of A inside the B-band is exactly $P(A \mid B)$. The cut runs straight precisely when

$$P(A \mid B) \;=\; P(A)$$

— conditioning on B changes nothing about A. On $P(B) > 0$ this is an equivalent characterization of independence, and the whole previous chapter re-reads through it. The die jewel: $P(\text{even} \mid \{1,2,3,4\}) = \frac{2}{4} = \frac{1}{2} = P(\text{even})$ — independent, the step flattens. One pip over: $P(\text{even} \mid \{1,2,3\}) = \frac{1}{3} \neq \frac{1}{2}$ — the step is back. Mutually exclusive events re-read as $P(A \mid B) = 0$: maximal information, the strongest step there is.

Two cautions keep the picture honest. The product form stays the *definition* — it survives $P(B) = 0$ and treats A and B symmetrically, which the conditional form does not. And conditioning never mutates the original measure: after the recount, $P(A)$ still answers every question about the old square. Both measures remain available; "given B" only says which one a question is asking.

5.3 The multiplication rule

Read the definition backwards and it becomes a way to *build* joint probabilities:

$$P(A \cap B) \;=\; P(B)\, P(A \mid B).$$

On the square this is one rectangle — a width times a conditional height — and it is the counting chapter's rule of product carrying probabilities: a "souped up rule of product" [42]. It is also the license the independence chapter asserted without showing: the aces' 1/221 was $\frac{4}{52} \cdot \frac{3}{51}$ all along — $P(A_1)\, P(A_2 \mid A_1)$, a probability times a conditional probability. The shrinking pool was never lawless; it was conditional, and now the law is on the page. Chaining the same move expands any intersection — $P(A_1 \cap \cdots \cap A_n) = P(A_1)\, P(A_2 \mid A_1) \cdots P(A_n \mid A_1 \cap \cdots \cap A_{n-1})$ — and since every ordering of the events is a valid expansion, the chain rule is "$n!$ theorems in one" [9]. One more habit falls out: time reversal costs nothing. For two cards, $P(S_1 \mid S_2) = 12/51 = P(S_2 \mid S_1)$ — conditioning is *re-measuring* a space of outcomes, not re-running the experiment, so "the second card was a spade" restricts the square as legally as the first.

5.4 Total probability, and trees

Slice the square into columns by a partition $B_1, \ldots, B_n$ — every outcome in exactly one column. Inside column i, the event A is a rectangle of width $P(B_i)$ and height $P(A \mid B_i)$; the rectangles tile A, so adding them up is the law of total probability:

$$P(A) \;=\; \sum_i P(B_i)\, P(A \mid B_i)$$

— an unconditional probability is the weighted average of its conditional pieces. On the die, partition by the jewel's own band: $P(\text{even}) = \frac{1}{3} \cdot \frac{1}{2} + \frac{2}{3} \cdot \frac{1}{2} = \frac{1}{2}$, the two columns' rectangles adding back to the whole. A tree diagram is this same square drawn as branches: edges carry conditional probabilities, leaves are intersections priced by the multiplication rule, and summing the circled leaves is the same addition — the tree is notation, not new mathematics. The alignment chapter's trellis has already run this law many times over: each forward column $\alpha_t(s)$ is total probability over the predecessor states — the partition is "which state the path came from", and the recurrence adds up the rectangles.

5.5 Two slices, one square

$P(A \mid B)$ and $P(B \mid A)$ share their numerator — the same overlap $P(A \cap B)$ — and share nothing else: one divides by the B-band, the other by the A-band [3]. Confusing them is the inversion fallacy, and an exact hit shows how far apart they can sit: over five coin flips, $P(\text{first is H} \mid \text{five heads}) = 1$ while $P(\text{five heads} \mid \text{first is H}) = \frac{1}{16}$.

The centerpiece is a pair of diagnoses. One test — it catches 9 in 10 sick people, and falsely flags 1 in 10 healthy ones — read on two populations, in whole-person counts. At 10% prevalence, a 100-person cohort yields 9 sick positives and 9 healthy ones: $P(\text{sick} \mid +) = 9/18 = \frac{1}{2}$. At 1% prevalence, a 1000-person cohort yields 9 sick positives and 99 healthy ones: $P(\text{sick} \mid +) = 9/108 = 1/12$. Same test, same overlap read the same way — but the prior travels inside the counts, and the counts keep the two slices from blurring together [10]. The series leaves on the doorstep of something bigger: multiply out the two definitions and the shared numerator gives the identity $P(A)\,P(B \mid A) = P(B)\,P(A \mid B)$ — one visible step from turning a known $P(B \mid A)$ into a wanted $P(A \mid B)$.

5.6 What exactly you condition on

The tool's last lesson is about its input. A family has two children; you learn there is *at least one girl*. Among the four equally likely family chips (BB, BG, GB, GG), three have a girl and one of those is two girls: $\frac{1}{3}$. But suppose what actually happened is that one child ran through the room and it was a girl. Families with two girls were twice as likely to produce that sighting as mixed families — draw the announcement rule as weights and the answer moves to $\frac{1}{2}$ [54]. Both computations are correct conditionings — on *different* events. The conditioning event includes *how you learned what you learned*.

And independence itself splits in two under conditioning. Take the two trick coins — one lands heads 9 times in 10, the other 1 time in 10 — pick one at random and flip it twice. Marginally the flips are dependent: $P(\text{both heads}) = 41/100 \neq \frac{1}{4}$ — the first head is evidence about which coin you hold, so it raises the odds of a second. Yet *given the coin*, the flips factor exactly [55]. Conditional independence is a third thing — neither implied by independence nor implying it — and it is the independence chapter's common-cause beat made quantitative: the coin is the common cause, and conditioning on it switches the dependence off. This is also the alignment chapter's assumption, finally stated in full: CTC's frames are not asserted independent outright, but independent *given the input* — condition on the audio, and the per-frame product becomes exactly the licensed chain.

Where this goes. The identity this chapter left on the doorstep — $P(A)\,P(B \mid A) = P(B)\,P(A \mid B)$ — is Bayes' front door, and the next chapter walks through it: one division, and the inversion fallacy's two slices become a lawful update rule. The two-children lesson scales up there too — Monty Hall, deferred here on purpose, is ordinary conditioning once the host's *protocol* is part of what you condition on. And several chapters on, when products of many conditional factors start underflowing, logarithms will turn this chapter's multiplications into additions.

Practice problems

Problem 5.1. Three fair coins are flipped. What is the probability all three landed heads — before any news, and then given that the first coin landed heads?

Hint: Recount: how many outcomes survive the news, and how many of those are all-heads?

Problem 5.2. Two cards are drawn without replacement. Use the multiplication rule to price $P(\text{both aces})$, identifying each factor as a probability or a conditional probability.

Hint: $P(A_1 \cap A_2) = P(A_1)\,P(A_2 \mid A_1)$; what deck does the second draw face?

Problem 5.3. An urn holds 5 red and 2 blue balls; two are drawn without replacement. Compute $P(\text{second is red})$ by total probability over the first draw, and compare it with $P(\text{first is red})$.

Hint: Partition on the first draw's colour; two rectangles.

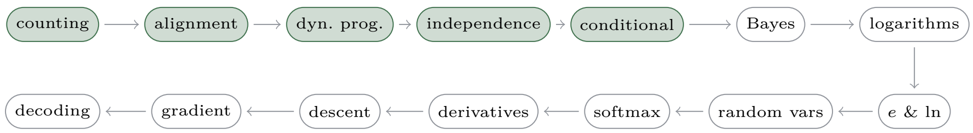
Problem 5.4. A test catches 9 in 10 sick people and falsely flags 1 in 10 healthy people. Compute $P(\text{sick} \mid +)$ in whole-person counts for a 100-person cohort at 10% prevalence, and for a 1000-person cohort at 1% prevalence.

Hint: Count the sick positives and the healthy positives separately, then take the share.

Problem 5.5. [stretch] One of two coins is chosen by a fair flip: one lands heads with probability $\frac{9}{10}$, the other $\frac{1}{10}$. The chosen coin is flipped twice. Show the flips are dependent, but conditionally independent given the coin.

Hint: Compute $P(H_1 \cap H_2)$ by total probability over the coin; compare with $P(H_1)P(H_2)$; then redo the check inside each coin's world.

The road so far



Conditioning ended one step short on purpose: the shared-numerator identity stood on the page with nothing asked of it. The next chapter asks the question every diagnostic test asks — you saw the evidence; which way does the inference run? — and the answer is a division so short it barely deserves its fame. What deserves the fame is what the division does under iteration.

Chapter 6

Bayes' rule

The conditional chapter ended one visible step short of something: the shared-numerator identity $P(A)\,P(B \mid A) = P(B)\,P(A \mid B)$, standing on the page with nothing yet asked of it. This chapter asks. One division turns the identity into the law of updating beliefs — and then the real work begins, because the formula is the least useful form of the law. Counts compute it, odds compress it, and a game-show host stress-tests what it actually takes as input.

6.1 Through the front door

Divide the standing identity by $P(B)$ — positivity inherited from the conditional chapter's definitions — and Bayes' rule is already on the page:

$$P(A \mid B) \;=\; \frac{P(B \mid A)\,P(A)}{P(B)}.$$

Nothing new was proved; the division is the two-slices picture re-read, and the denominator is the conditional chapter's law of total probability run over the hypothesis columns. What is new is the *reading*, and the three names that come with it: $P(A)$ is the *prior* (what the hypothesis was worth before the news), $P(B \mid A)$ is the *likelihood* (how well the hypothesis explains the news), $P(A \mid B)$ is the *posterior* (what it is worth after). The claim of the whole chapter, in one line: evidence does not *determine* beliefs, it *updates* them — posterior $\propto$ prior $\times$ likelihood, normalize last.

6.2 Counting it out

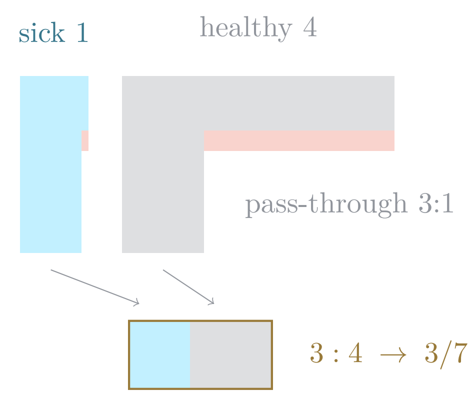
The first real computation needs no formula at all. Diseasitis, in whole students: a 100-student cohort, 20 sick and 80 healthy. The test flags 18 of the sick and 24 of the healthy. A student tests positive; the probability they are sick is the share of positives who are sick:

$$\frac{18}{18+24} = 3/7.$$

That is Bayes' rule, computed by counting — the prior never appeared because it travelled *inside* the counts (20 sick students out of 100 is the prior, already multiplied in). This is why natural frequencies cure base-rate neglect where the probability format fails [14]: posed in probabilities, the problem is solved by 4% → 24% of laypeople — and reformatting alone moves physicians from 21% → 87%. When you must *communicate* a posterior, hand over the counts.

6.3 The odds form

Now compress. Keep only the *ratio* between the two hypotheses and watch what the evidence does to it — the waterfall:



Two streams pour down at the prior widths, 1 sick to 4 healthy. The test lets 9 in 10 of the sick stream through and 3 in 10 of the healthy stream — pass-through in ratio 3:1 — and the pool at the bottom collects 18:24, which is 3 : 4: the same 3/7 as the counted cohort. The waterfall is the cohort chips, and the tree, and the two slices, drawn as one object with renormalization deferred to the very end — and its law is a single multiplication [62]:

$$\text{posterior odds} \;=\; \text{prior odds} \times \text{likelihood ratio}.$$

Only ratios matter: scale either stream's width or both pass-through fractions by any common factor and the pool's ratio never moves. One discipline comes with the compression — odds and probabilities answer different questions, so keep $3:4$ and $3/7$ visibly distinct.

6.4 One test, two patients

The odds form factors the conditional chapter's prevalence pair into its parts. That test — 9 in 10 sensitive, 1 in 10 false positives — has *one* number of its own: the likelihood ratio 9 [2]. The posterior belongs to the patient:

$$1{:}9 \times 9 = 1{:}1 \;\to\; \tfrac{1}{2}, \qquad 1{:}99 \times 9 = 1:11 \;\to\; 1/12.$$

The counted 9/18 and 9/108 of the conditional chapter fall out as derivations instead of tallies — same answers, one multiplication each. And the marketing phrase "90% accurate" stands revealed as one word hiding two numbers (sensitivity and false positive rate), being silently read as a third (the posterior). A posterior can never be stated without its prior; a test's quality can never be stated as one number that *is* a posterior.

6.5 Yesterday's posterior

Evidence arrives in sequences, and the odds form makes iteration effortless: yesterday's posterior is today's prior. On the conditional chapter's own trick coins — heads 9 in 10 versus 1 in 10, so each head carries likelihood ratio 9 — start at 1:1: one head moves the odds to 9:1, a second to $81:1$; a head *then a tail* lands back at exactly 1:1 [41]. That exact cancellation is the point: if evidence *replaced* belief, the order and mix of observations could not wash out; because evidence *reweights* belief, a tail's ratio $\frac{1}{9}$ undoes a head's 9 automatically. Two captions guard the habit. Likelihood ratios multiply only when the observations are conditionally independent *given the hypothesis* — the conditional chapter's closing lesson, cashed at the exact moment it is needed (it is the disease being common cause of both test results that makes the second test still informative, yet lets its ratio simply multiply on). And a prior of zero stays zero under any evidence of nonzero probability — no finite likelihood ratio rescues a hypothesis that was never in the running.

6.6 The host's protocol

Monty Hall, at last — deferred by the conditional chapter until the tools could handle it honestly. You pick door 1; the host, who knows where the car is, opens door 3 on a goat; switch to door 2? The trap is conditioning on the revealed *fact* ("door 3 has a goat") instead of the host's *action* under his protocol — the two-children lesson at series scale. Put a uniform prior on the car's position and let the likelihood be the probability of the action *the host actually took*, one row per protocol [49]: the standard host (never the car, coin-flip when free) assigns the opening likelihoods $(\frac{1}{2}, 1, 0)$ across the three car positions, and switching wins with probability 2/3; Monty Fall, who slipped and opened a random door that merely *happened* to hide a goat, assigns $(\frac{1}{2}, \frac{1}{2}, 0)$ — and switching wins only 1/2; Monty Crawl, who opens the lowest-numbered door he can, was *forced* high in opening door 3 — and switching wins with certainty. Same door, same goat, three answers: the information is not in what was revealed but in what the protocol made that revelation *cost*. Rosenthal's proportionality principle — posterior $\propto$ prior $\times$ likelihood of what was observed — is just Bayes with the uniform prior left implicit, and it is the series' closing rule: *condition on what happened, the way it happened — then multiply.*

Where this goes. The multiplications of this chapter are about to become additions: the logarithm chapters build a ruler on which a likelihood ratio is a fixed *distance*, and iterated updating is walking — with a zero prior sitting infinitely far down the scale, which is why no finite evidence reaches it. The likelihood, named here on the front door, returns as the lens the softmax chapter reads per-frame distributions through. And at the road's end, the decoding chapter cashes the independence chapter's common-cause warning the Bayesian way: an external language model is a prior, multiplied onto per-frame likelihoods — condition on the way the audio happened, then multiply.

Practice problems

Problem 6.1. In the Diseasitis cohort — 100 students, 20 sick of whom the test flags 9 in 10, 80 healthy of whom it falsely flags 3 in 10 — a student tests positive. Compute the probability they are sick using whole-person counts only.

Hint: How many sick positives; how many positives in all?

Problem 6.2. Recompute the Diseasitis posterior in the odds form, and check it against the counts.

Hint: Prior odds from 20:80; the likelihood ratio from the two flag rates.

Problem 6.3. A test has sensitivity $\frac{9}{10}$ and false positive rate $\frac{1}{10}$. Using the odds form, find $P(\text{sick} \mid +)$ for a patient from a population at 10% prevalence, and for one at 1%.

Hint: The test contributes one number; only the prior odds change.

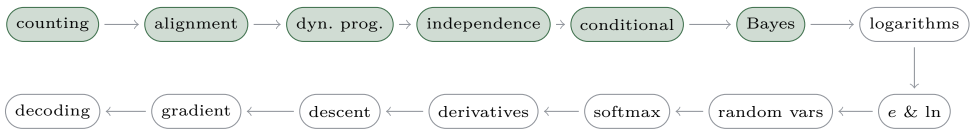
Problem 6.4. One of two coins (heads $\frac{9}{10}$ versus $\frac{1}{10}$) is chosen by a fair flip. Track the odds that you hold the heads-heavy coin after observing H, after HH, and after HT — and give each as a probability.

Hint: One likelihood ratio per flip; multiply. What is a tail's ratio?

Problem 6.5. [stretch] You pick door 1; the host opens door 3, showing a goat. Compute the probability that switching to door 2 wins under each protocol: (a) the standard host — never opens your door or the car's, coin-flips when both goat doors are available; (b) Monty Fall — opened a random door of the other two and happened to reveal a goat; (c) Monty Crawl — always opens the lowest-numbered door he legally can.

Hint: Uniform prior over the car's position; the likelihood is the probability of *this host* opening door 3, given each car position.

The road so far



Bayes' ladder multiplies: each round of evidence scales the odds again. Multiplying is exactly what the trellis does too — and you have already seen where repeated multiplication leads: down. The next chapter builds the instrument that turns multiplying into adding — the counting strip, the evidence ruler — and walks to the cliff-edge where floating point dies and the log-add identity saves the trellis's additions.

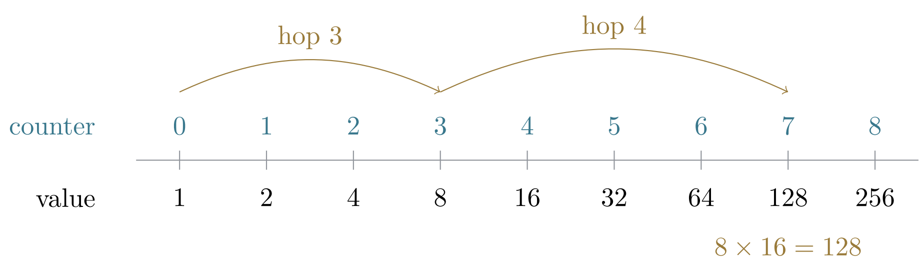
Chapter 7

Logarithms

Everything on this road multiplies. The independence chapter multiplied one probability per trial; the conditional chapter multiplied a width by a height; the Bayes chapter multiplied prior odds by a likelihood ratio; and the trellis multiplies a per-frame score for every frame of every path. This chapter is the tool that turns all of that multiplying into adding — not by a trick, but by a reading: a logarithm *counts multiplicative steps*.

7.1 The counting strip

Write two rows. The bottom row starts at 1 and doubles: $1, 2, 4, 8, \ldots, 256$. The top row just counts the doublings: $0, 1, 2, \ldots, 8$. That strip is the whole chapter's spine. "2 to the what is 64?" is answered by reading the counter above 64 — six. A logarithm inverts the *question*, not the graph of some function; no curve is needed to say what it is.



The base is the stride of one step, and the strip hands over the definition's conditions for free [45]: base 1 never moves (1's ladder reads $1, 1, 1, \ldots$ — no counter is the answer, or every counter is), and a positive base never leaves the positives, so only $x > 0$ has a counter at all. Only then, the formula:

$$\log_b x = y \iff b^y = x, \qquad b > 0,\ b \neq 1,\ x > 0.$$

7.2 One fact, three notations

$2^6 = 64$, $\log_2 64 = 6$ and $\sqrt[6]{64} = 2$ are not three facts. They are one fact about the triple $(2, 6, 64)$, asked from three corners [4]: cover the value and a power asks for it; cover the exponent and a logarithm asks for it; cover the base and a root asks for it. In particular $b^{\log_b x} = x$ is the same number occupying two corners at once — an *undo*, never a “cancellation”. Students taught that logs “just disappear” reliably extend the disappearing act to places it does not go [34].

The best-documented place it does not go is addition. In base 10, $\log_{10}(10 + 10) = \log_{10} 20 \approx 1.301$ — not $1 + 1 = 2$. There is no law for the log of a sum. The trap earns an extra sentence because the base-2 instance is coincidentally *true*: $\log_2(2+2) = 2$ exactly, but only because $2+2$ happens to equal 2×2 — one lucky check away from a wrong rule.

7.3 Multiplying is adding

Multiply 8×16 on the strip. 8 sits at counter 3 and 16 at counter 4, so their product is reached by hopping 3 then 4 more — counter 7, value 128, as the figure’s arcs walk it. Multiplying values *is* adding counters:

$$\log_b(xy) = \log_b x + \log_b y.$$

The law was once hardware: two logarithmic scales sliding against each other (Gunter’s single scale, 1620; Oughtred’s pair, c. 1622) add lengths so that their readings multiply [61].

Change of base is a change of *stride*. Count 64 in strides of 4 instead of 2: each 4-stride is two 2-strides, so $\log_4 64 = 6/2 = 3$ — the base is a unit of measurement, nothing more. Unit conversion also explains a number worth keeping: ten doublings give $1024 \approx 1000$, about three tenfoldings, so one doubling is worth about three tenths of a digit — $\log_{10} 2 \approx 0.301$.

7.4 Shrink counts

The independence chapter halved a unit square four times and found the cell $HHTH$ with area $(1/2)^4 = 1/16$. Read on the strip, that cell sits four steps *below* 1: $-\log_2 \frac{1}{16} = 4$. Probabilities in $(0, 1)$ have negative logs for exactly this reason — they are counts of *shrinkings*, and the sign only records which direction the strip was walked. pH is the everyday instance: acidity 7 is a concentration tenfolded down seven times.

Two boundary readings matter later. First, $\log 0 = -\infty$: the Bayes chapter's zero prior sits infinitely far down the strip — the log-space form of "no finite evidence rescues it". Second, slow is not bounded: name any N and 2^N sits on the strip. Logarithms grow without limit, just very politely.

7.5 The evidence ruler

The Bayes chapter's coins multiplied: likelihood ratio 9 per head, odds $1 \to 9 \to 81$. Re-plot that ladder on a base-3 ruler. Since $9 = 3^2$, each head adds exactly 2: the marker steps $0 \to 2 \to 4$. Each tail subtracts the same length, so H, H, T, T walks it back to exactly 0. "Each head multiplies by 9" and "each head adds the same length" are the same data — the multiplicative waterfall and the additive ruler are one object. This is evidence as *weight*: Turing scaled such a ruler in base $10^{1/10}$ and called its unit the deciban, roughly the smallest weight of evidence a human can feel [15].

7.6 The underflow cliff

Now the payoff the alignment chapters have been waiting for. A machine multiplying many small probabilities is walking down the strip, and the floor is closer than it looks: float64 bottoms out at 2^{-1074}, so 0.1^{323} still survives — barely, as a subnormal — while 0.1^{324} is exactly 0.0; not small, *gone* [56]. Meanwhile the counter row walks on unharmed: the $\log_{10}$ sum of the same factors is exactly -324. In float32, the type networks actually train in, the same death arrives after only 46 frames of probability 0.1. Sums of logs survive what products of probabilities cannot.

One thing breaks in the move. The trellis multiplies scores *along* a path — under the log, products become sums, done — but it also *adds* the α's of merging paths, and log space has no cheap addition. The repair is one identity, the log-sum-exp move [58]:

$$\ln(a + b) \;=\; \ln a + \ln\!\big(1 + e^{\ln b - \ln a}\big), \qquad a \geq b.$$

Factor out the larger term and the leftover $e^{\ln b - \ln a}$ lives in $(0, 1]$ — finite, tame, no cliff. This is the one place the series writes ln and e, and the video prints its debt on screen: *names on loan from calculus*. This guide carries the same loan and repays it in the e-and-ln chapter ahead; until then, read ln as "the counter row in some fixed base" — every law used here is base-generic. One credit note, kept straight: log-space CTC is the 2012 book's move [18]; the 2006 paper rescaled instead.

Where this goes. The loan is repaid next: the e-and-ln chapter finds the base nature rules the strip in and re-reads this identity symbol by symbol. The shrink counts return averaged — the random-variables chapter finds the sixteen-cell square carries exactly 4 bits of surprisal on average. And the ruler returns twice more: differentiated, when the derivative toolkit turns the log-sum-exp ruler's sensitivities into softmax's shares, and walked end to end, when the gradient chapter runs the whole trellis in log space.

Practice problems

Problem 7.1. How many doublings take 1 to 512 — that is, what is $\log_2 512$?

Hint: Count hops on the strip; the ladder in the figure stops one doubling short.

Problem 7.2. A student writes $\log_{10}(100+1000) = \log_{10} 100+\log_{10} 1000 = 5$. What is the true value, roughly — and why is the base-2 near-miss $\log_2(2+2) = 2$ no excuse?

Hint: There is no law for the log of a sum; 1100 sits between 10^3 and 10^4.

Problem 7.3. A fair coin comes up heads ten times running, probability $(1/2)^{10} = 1/1024$. What is $-\log_2$ of that probability? And what is the pH-style reading of a concentration of 10^{-7}?

Hint: Count shrinkings — the strip walked downward.

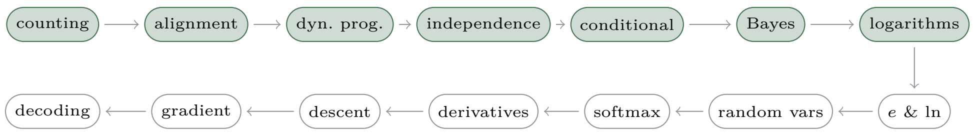
Problem 7.4. A test has likelihood ratio 9 per head (and 1/9 per tail). Starting from even odds, the sequence H, H, H, T arrives. Where does the marker sit on the base-3 evidence ruler, and what are the odds?

Hint: $9 = 3^2$: each head is $+2$ on the ruler, each tail -2.

Problem 7.5. [stretch] A model scores 324 frames at probability 0.1 each. In float64, what does the product return, and what does the counter row say? Then price a log-space *addition*: two merging paths each carry weight 2^{-10} — what is $\log_2$ of their sum, by the chapter's closing identity?

Hint: The product falls off the cliff; the sum of logs does not. For the addition, $a = b$ makes the identity's shifted term exactly 1.

The road so far



The strip works in any base — 2, 3, 10 — and the chapter closed on a loan: one identity written in ln, its names borrowed from calculus. The next chapter is calculus arriving to pay: the base nature picks, the counter row that was always secretly ruled in it, and the borrowed line re-read symbol by symbol until every mark on it means something.

Chapter 8

e and the natural logarithm

The logarithms chapter ended in debt. Its closing identity was written with two symbols — ln and e — whose names it admitted, on screen, were on loan from calculus. This chapter repays the loan. The counting strip's strides come in every size — base 2, base 3, base 10, each a legitimate unit — and what follows finds the one stride that is nature's own, landing ln not as a new function but as the counter row every earlier strip was secretly ruled in.

8.1 The split year

Jacob Bernoulli, 1683 [47]: one dollar, one year, 100% interest. Paid once, it doubles: 2. Paid in two half-year instalments, the midyear interest starts earning interest of its own: $(3/2)^2 = 9/4 = 2.25$. Each further split is more, smaller multiplicative hops — quarterly $(5/4)^4 = 625/256$, then monthly, weekly, hourly:

$$\left(1 + \tfrac{1}{n}\right)^n \;=\; 2.25,\; 2.4414,\; 2.6130,\; 2.7146,\; 2.7181, \ldots$$

Two wrong intuitions die against this table. "More compounding grows without bound" — no: the table is strictly increasing yet never reaches 3; Bernoulli proved the ceiling sits between 2 and 3, and never named it. "$1 + \frac{1}{n} \to 1$, so the power collapses to 1" — no: the shrinking hop is taken a growing number of times, and the two effects balance at a ceiling. Only after the crowding is watched does the formula get to summarise it:

$$\lim_{n\to\infty} \left(1 + \tfrac{1}{n}\right)^n = e$$

— the first number in history defined as a limit: crowded from below, named only decades later.

8.2 Zoom until straight

To ask "how fast does 2^x grow *here*?" the series builds exactly one piece of calculus, in three beats and no more. First: magnified enough, a smooth curve is indistinguishable from a straight line [12], so "the slope here" means something. Second: read the slope of 2^x at 0 as rise over run with dt a real number on screen — the ratio $(2^{dt}-1)/dt$ reads $1.0,\ 0.7177,\ 0.6956,\ 0.69339$ at $dt = 1, 0.1, 0.01, 0.001$, and the readout *settles*; the word "instantaneous" is never needed, and never used. Third: one strip law carries the readout everywhere — $2^{x+dt} = 2^x \cdot 2^{dt}$, a hop is the same length wherever it starts — so the slope at any x is the height there times the slope at 0:

$$\text{slope of } 2^x \text{ at } x \;=\; 2^x \times 0.6931\ldots$$

Growth rate proportional to amount: that is what *exponential* means.

8.3 The mystery constants

Line up the settling constants for several bases [5]: base 2 $\to$ 0.6931, base 3 $\to$ 1.0986, base 8 $\to$ 2.0794. And 2.0794 is exactly three of 0.6931 — because $8 = 2^3$ is three base-2 strides. The constants obey the counting strip's laws *before anyone has named them*: they are stride lengths, measured in some undisclosed unit. Then the squeeze: $0.6931 < 1 < 1.0986$ — between base 2 and base 3 sits the base whose constant is exactly 1, the base whose growth rate *is* its height [44]. Its settling rows read $1.0517,\ 1.0050,\ 1.0005 \to 1$, and its value,

$$e = 2.718281828459\ldots,$$

is the split year's ceiling, back from a different question. Note what e is not: big. 3^x outruns e^x easily — e is the *self-paced* base, not the fast one.

8.4 The natural stride

Disclose the unit. The slope-at-0 constant of base b is the length of one base-b stride measured in e-strides — change of base, which the logarithms chapter already taught, applied once. So the constants are $\log_e b$, written $\ln b$, and the strip returns ruled in nature's units: values $1, 2, e, 4, 8$ under counters $0,\ 0.693,\ 1,\ 1.386,\ 2.079$. ln is not a harder animal — it is the counter row in natural units, and every log ever met was ln in a stretched unit.

Two small facts close the circle the popular treatments leave open [52]. A tiny hop costs its own size: multiplying by 1.01 costs 0.0099503 natural strides — $\ln(1+x) \approx x$ for small x. Therefore

$$n \ln\Big(1+\tfrac{1}{n}\Big) = 0.9531,\ 0.99503,\ 0.99950\ \to\ 1:$$

the compounding ceiling is the value whose natural counter is exactly 1 — the limit-e of the split year and the slope-e of the mystery constants meet, shown on the strip. (That an increasing, bounded table has a limit at all is analysis — named here, not derived.)

8.5 Rate times time

Growth has one dial: e^{rt}, with e as its unit. The knobs only multiply — ten years at 3% is one year at 30%. Doubling means $rt = \ln 2$, so $t \approx 0.693/r$: at 5%, 13.86 years. (100×0.693 is the mathematics; the popular 72 is a divisibility convention — friendlier to mental division, not truer.) And continuous compounding closes Bernoulli's ledger: daily compounding at 5% lands within 3.6×10^{-6} of $e^{0.05}$. The limit is not an abstraction; it is where the daily ledger already stands.

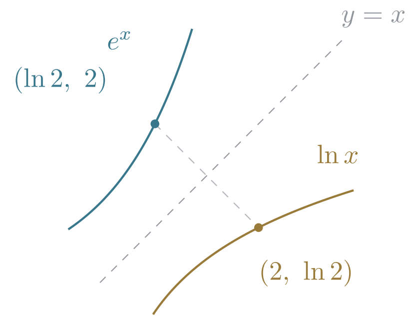
8.6 The debt repaid

Now the identity, symbol by symbol:

$$\ln(a+b)\ =\ \ln a + \ln\!\big(1 + e^{\ln b - \ln a}\big), \qquad a \geq b.$$

$\ln a$ is the natural counter of a. $e^{(\cdot)}$ is undo, never cancel — the logarithms chapter's own grammar. And $a \geq b$ factors the larger term out so the shifted term $e^{\ln b - \ln a}$ lives in $(0,1]$: finite, tame. At the cliff's own scale the details are the whole game: in float64, $1 + e^{-40}$ is exactly 1.0, so the naive route computes $\ln 1.0 = 0.0$ — the smaller term silently deleted — while $\text{log1p}(e^{-40}) \approx 4.248 \times 10^{-18}$ survives, because $\ln(1+x) \approx x$ means handing x straight to the log keeps it. The identity the logarithms chapter used on credit now has all its parts.

Then, last, the payoff the algebra series explicitly deferred: mark $\ln 2 = 0.693$ on the e^x graph — the input that yields 2 — one point at a time, and only then flip the whole curve across $y = x$:



The inverse graph arrives as something earned — never as the definition, a starting point documented to misfire. Euler gave the number its letter in a 1731 letter, put it in print in 1736, and the *Introductio* of 1748 carries it to 18 places [27]. The series' takeaway box says it in one line: *e is the base whose stride is* 1 *— nature's unit for growth.*

Where this goes. The ceiling returns almost immediately, mirrored: the random-variables chapter finds zero successes in n trials of chance $1/n$ crowding $1/e$ from below. The softmax chapter answers "why e" in every probability machine with this chapter's base change — every base above 1 is a temperature; e is merely the one whose counter is natural. And the derivative toolkit turns the mystery constants into what they always were — derivatives — with $\frac{d}{dx}e^x = e^x$ as the kit's crown.

Practice problems

Problem 8.1. Compute the split-year table's first two refinements exactly, as fractions: $(1 + \frac{1}{2})^2$ and $(1 + \frac{1}{4})^4$. Verify that the second exceeds the first and that both sit below 3.
Hint: Powers of small fractions; nothing here needs a decimal until the final comparison.

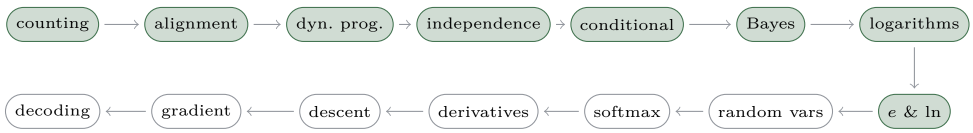
Problem 8.2. Read the slope of 2^x at 0 the series' way: compute $(2^{dt} - 1)/dt$ at $dt = 0.1$, 0.01 and 0.001. What is the ratio settling toward?
Hint: Rise over run with dt a real number — no limit notation required.

Problem 8.3. Base 8's mystery constant is exactly three of base 2's, because $8 = 2^3$. Without any new measurement, what is base 32's constant?
Hint: $32 = 2^5$: five strides.

Problem 8.4. An account grows continuously at 2% per year. How long until it doubles? What would the popular rule of 72 say, and which answer is the mathematics?
Hint: Doubling means $rt = \ln 2$.

Problem 8.5. [stretch] Compute $\ln(e^{-1000} + e^{-1001})$. What happens on the naive route through value space, and what does the repaid identity give?
Hint: e^{-1000} does not exist in float64. Factor out the larger term.

The road so far



Between e and the trellis stands one more block: the measurements themselves. The square from the independence chapter is about to be stamped — every cell inked with a number before anything is rolled — and sorted into the columns that make a distribution. Expectation arrives as a balance point, and the ceiling from last chapter returns mirrored: zero successes in n trials of chance 1/n, crowding
anchor007.oneovere from below.

Chapter 9

Random variables

So far the road weighs *events*: regions of a square, shares of area. But the questions it is heading toward are about *numbers* read off outcomes — how many heads, which symbol, how large a score. A random variable attaches a number to every outcome; this chapter builds the attachment, watches it sort into a distribution, and finds the one number to act on.

9.1 The stamped square

A die is usually introduced as a set of six outcomes. This series opens differently: the die as a *function* — $X(\text{face})$ = the number shown — ink stamped on the sample space before anything is rolled [30]. Then the square the independence chapter quartered twice returns for its third appearance: sixteen equal cells, one per four-flip sequence, each stamped with its head count, 0 through 4. The stamp is fixed; the only random object is where the dart lands; the label is *looked up*, never generated. Only then, the formula:

$$X : \Omega \to \mathbb{R}.$$

Every measurement attached to a random process is this — a fixed rule reading a random outcome.

9.2 Sort the square

Slide the stamped cells into columns grouped by their stamp. Area is conserved — nothing is created, nothing weighed twice — and the bars that form are the distribution: five *unequal* bars born from sixteen *equally likely* cells,

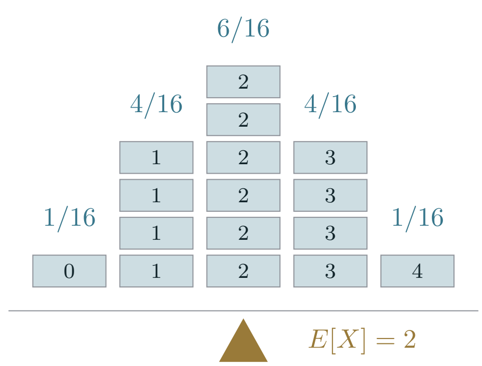
$$P(X = k) \;=\; \frac{\#\{\omega : X(\omega) = k\}}{16} \;=\; (1, 4, 6, 4, 1)/16.$$

The unequal weights are visible counts of equal cells — which is the working refutation of the equiprobability bias, the documented habit of reading randomness as uniformity that survives instruction unless the mechanism is shown [13].

Then the blueprint-versus-house beat: stamp the *same* square with $Y =$ tails. The sorted columns are identical — same pmf — yet $X \neq Y$, and $X + Y = 4$ in every single cell. The distribution forgets which cell was which; the variable remembers. Mistaking one for the other is the canonical error this scene exists to prevent.

9.3 The balance point

Expectation is defined, not simulated. Hang the die's six equal bars on a beam and ask where the fulcrum goes: it balances at 3.5 — which is not a face. "The point at which the distribution would balance" is the picture; a weighted average need not be attainable [46].



For the flips, the two ways of summing are shown equal once, and the sort is the proof: summing stamps over cells gives $32/16 = 2$, and summing values times column weights gives 2 — the sort was a *regrouping*, which is why

$$E[X] \;=\; \sum_x x \cdot P(X = x)$$

may be computed cell by cell or column by column. And the balance point belongs to the *measure*, exactly as independence did: put the independence chapter's biased die on the beam — double weight on the six — and the fulcrum moves to 27/7, again not a face. The level-three reading is Huygens, 1657, the first printed probability work: expectation is the fair price of the ticket — the number you act on, with no long run in sight [25].

9.4 Same outcomes add

$E[X+Y]$ is one sum over the *same* outcomes, and addition distributes — so

$$E[X+Y] \; = \; E[X] + E[Y]$$

with independence nowhere in the hypothesis: as Grinstead and Snell put it, mutual independence of the summands was not needed [20]. The demonstration runs at the dependent extreme: X and $4-X$ are as dependent as two variables can be, and their sum is 4, always. Then the independence chapter's 6×6 grid is re-read: the two-dice sum paints diagonals, distribution $(1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1)/36$, and $E = 7$ arrives twice — once by the diagonals' weighted sum, once as $3.5 + 3.5$. Linearity is the workhorse: it prices any bundle from its parts.

9.5 The binomial columns

Now the road's oldest promise closes. Count cells per column — the numerators of $(1, 4, 6, 4, 1)/16$ — and notice that counting four-letter H/T words with k heads is exactly the counting chapter's $\binom{4}{k}$. The coefficient is a *cell count*, never a memorized factor.

Re-cut the square with the cuts at $p = 1/4$. The cuts stay straight — straight cuts are the square's grammar for independence, and independence is unchanged; only the cut positions move. The cells go unequal, but every k-head cell keeps the *same* area $p^k q^{4-k}$, because a product does not care about factor order. So column k weighs (cell count) $\times$ (one cell's area), and the pmf assembles with nothing smuggled:

$$P(X=k) \; = \; \binom{n}{k} p^k (1-p)^{n-k},$$

at $p = 1/4$ concretely $(81, 108, 54, 12, 1)/256$, each one-head cell exactly $27/256$ with $\binom{4}{1} = 4$ such cells. Expectation costs no combinatorics at all: stamp each trial's indicator and add four expectations of p — $E = np$, here 1. The model's conditions are named, not assumed [24]: fixed n, two outcomes, constant p, independent trials — and the independence chapter's aces already marked the boundary: no replacement means no binomial. One more door opens here: zero successes in n trials of chance $1/n$ crowds $1/e \approx 0.3679$ from below — the previous chapter's ceiling, mirrored in a shrinking power.

9.6 Proportions converge

The closer turns the swamping intuition into exact numbers — exact binomial sums, no new machinery. Within $\pm 5\%$ of half the flips: the probability *climbs*, $0.4966 \to 0.7287 \to 0.9986$ at $n = 20, 100, 1000$. Within ± 5 *heads* of half: it *falls*, $0.7287 \to 0.2720 \to 0.0876$ at $n = 100, 1000, 10000$. The two $n = 100$ entries are the same band of outcomes, 45 to 55 — one number telling two stories. Proportions converge while counts spread: the gambler's fallacy dies by two columns moving in opposite directions. Chance *swamps*; it does not compensate.

The theorem these tables compute instances of is the weak law of large numbers — Bernoulli had it proved by about 1689; it reached print in 1713 [19, 26] — named here, and promised for the day variance arrives. One small closing count: average the logarithms chapter's surprisal over the sixteen equal cells and it comes to exactly 4 bits — a number waiting for the information-theory road.

Where this goes. The next move pins the data and sweeps the parameter: the likelihood lens that opens the softmax chapter reads a binomial table along the other axis, and this chapter's columns are that table. The per-frame distributions the trellis multiplies are pmfs exactly like these — the gradient chapter differentiates straight through them. And the weak law returns as a theorem when spread gets its own machinery: variance and the proof arrive together.

Practice problems

Problem 9.1. Three fair flips, X = number of heads. Stamp the eight cells and sort: what is the pmf of X?

Hint: Eight equal cells; group by stamp and count.

Problem 9.2. Where does the fulcrum balance for a fair die? And for the biased die with double weight on the six (weights $1, 1, 1, 1, 1, 2$ over 7)?

Hint: $E[X] = \sum x \cdot P(X = x)$, exact fractions throughout.

Problem 9.3. Two fair dice, S = their sum. Compute $E[S]$ twice: once from the distribution of S, once with no distribution at all.

Hint: The 6×6 grid's diagonals carry the counts $(1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1)$; linearity needs neither them nor independence.

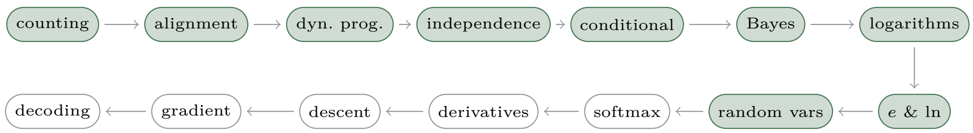
Problem 9.4. Assemble the binomial pmf for $n = 4$, $p = 1/4$ the sorted-square way: cell counts times one cell's area. Then check $E = np$ by the definition sum.

Hint: The counts are $1, 4, 6, 4, 1$; a k-head cell has area $(1/4)^k(3/4)^{4-k}$.

Problem 9.5. [stretch] Zero successes in n trials of chance $1/n$: compute $(1-1/10)^{10}$ exactly and $(1-1/100)^{100}$ to six places, and compare both with $1/e$. From which side, and in what order, do they approach it?

Hint: $(9/10)^{10}$ terminates as a decimal; the last chapter named the limit.

The road so far



A pmf answers: fixed model, which outcome? The per-frame matrices the trellis multiplies are pmfs exactly like the columns you just built. But training asks the transposed question — fixed outcome, which model? — and the next chapter turns the binomial table sideways to answer it, then builds the machine that turns arbitrary scores into per-frame distributions and the loss that scores the machine.

Chapter 10

Softmax and likelihood

The alignment chapter borrowed one thing on credit: each frame carries "the model's per-frame probability" for each symbol. The chapters since have built distributions honestly — regions of a square, sorted columns, balance points — and the random-variables chapter closed with a promise: *likelihood is next*. This chapter keeps that promise and pays the alignment chapter's debt in the same breath. It has two arcs and a join: first the *likelihood* arc — no new number, just an owned table read along its other axis — then the *softmax* arc — the machine that manufactures per-frame distributions from raw scores — and finally the loss that scores the machine, which is the exact quantity the road's last chapters differentiate.

10.1 One table, two questions

Take the binomial table the random-variables chapter sorted out of the square: $n = 4$ coin flips, the count of heads k down the rows, and now three candidate coins across the columns — $p = 1/4$, $1/2$, $3/4$. Every entry is one number, $P(k \mid p)$, written over the common denominator 256:

	$p = 1/4$	$p = 1/2$	$p = 3/4$
$k = 0$	81/256	16/256	1/256
$k = 1$	108/256	64/256	12/256
$k = 2$	54/256	96/256	54/256
$k = 3$	12/256	64/256	108/256
$k = 4$	1/256	16/256	81/256

Read a *column*: pin the coin, sweep the data. Each column is a pmf — the sorted-square bars — and sums to exactly 1. Now read a *row*: pin the *observed* data — say the flips came up $k = 3$ heads — and sweep the coin. The highlighted row, in the count convention $L(p) = \binom{4}{3}p^3(1-p)$, is $(3/64,\ 1/4,\ 27/64)$ across the three columns, and it sums to $23/32$ — visibly not 1. A pmf one way, not one the other. The row is a new *function of the parameter*, and it has a name: the likelihood. Fisher named it in 1921 and defined it in general in 1922 [31]. Same table, two questions — "given the coin, what data?" runs down; "given the data, which coin?" runs across. The formula, last:

$$L(p) \;=\; P(\text{data} \mid p), \qquad \text{data pinned, } p \text{ sweeping.}$$

10.2 The best explanation

The row's peak names the parameter that explains the data best. Discrete first: three die rolls come up $(6, 6, 3)$, and two candidate dice compete. The fair die assigns the sequence $(1/6)^3 = 1/216 \approx 0.0046$; the biased die the random-variables chapter built — double weight on six, $P(6) = 2/7$, the rest $1/7$ each — assigns $(2/7)^2(1/7) = 4/343 \approx 0.0117$. The biased die out-explains the fair one by the ratio $864/343 = 2.52$. That ratio is an *update factor* — a rung of the Bayes chapter's ladder, the number a prior would multiply — not a verdict; maximum likelihood is what remains of Bayes when the prior is flat.

Then the continuous sweep. For the coin data $k = 3$ in $n = 4$, trace $L(p) = 4p^3(1-p)$ over the whole interval: $3/64 \approx 0.0469$ at $p = 1/4$, exactly $1/4$ at $p = 1/2$, and the peak $27/64 \approx 0.4219$ at

$$\hat{p} \;=\; \tfrac{3}{4} \;=\; \tfrac{k}{n},$$

the observed proportion. This is the random-variables closer read backwards: the proportion converges to p, so the proportion is the best guess at p. One guard before moving on: the area under this curve is exactly $1/5$, not 1 — a likelihood is never a distribution over the parameter. Making it one takes a prior and a renormalization — the Bayes chapter's move, pointed at, not performed.

10.3 Log the likelihood: add to survive

Three reasons to take the log, in rising order of necessity. First, it changes nothing that matters: ln is monotone, so L and $\ln L$ peak at the same $\hat{p} = 3/4$. (The exact sequence HHTH, without the $\binom{4}{3}$ count, has likelihood

$p^3(1-p)$ — its log curve sits exactly $\ln 4 = 1.3863$ lower, same shape, same peak; the constant lift moves nothing, said once.)

Second, the log is the *native scale* of accumulating independent evidence. Five per-frame probabilities (0.9, 0.8, 0.7, 0.9, 0.6) multiply to 0.27216; their logs add to -1.3014, and exponentiating the sum recovers the product exactly. This is the counting strip from the logarithms chapter carrying likelihood now — multiply-is-add, with probability riding the strip — and it is why the evidence ruler was a log-likelihood ruler all along.

Third, survival. The logarithms chapter walked off the underflow cliff on purpose: forty-six factors of 0.1 store as exactly 0.0 in float32, and 0.1^{324} is exactly 0.0 even in float64. A trained model multiplies *hundreds* of per-frame factors — the product is dead on arrival. The sum of logs walks to -105.9189 and -324 unharmed. The losses of likelihood-trained models are log-likelihoods because the answer is untouched, the arithmetic becomes additive, and the additive form is the only one that survives the hardware.

10.4 The probability machine

Now the other arc. A model hands each frame raw scores — say $z = (2, 1, 0)$ — and the road needs a genuine distribution: positive shares, total 1. The naive repair, divide by the sum, fails twice on screen. On z itself it gives $(2/3,\ 1/3,\ 0)$ — the score-0 class erased outright; shift every score up by one, $(3, 2, 1)$, and the shares move to $(1/2,\ 1/3,\ 1/6)$ — but a uniform shift changed nothing about which score beats which; and a negative score would make a negative "probability". The repair that works does two things in order: *exponentiate* — every score lands positive — then *normalize*:

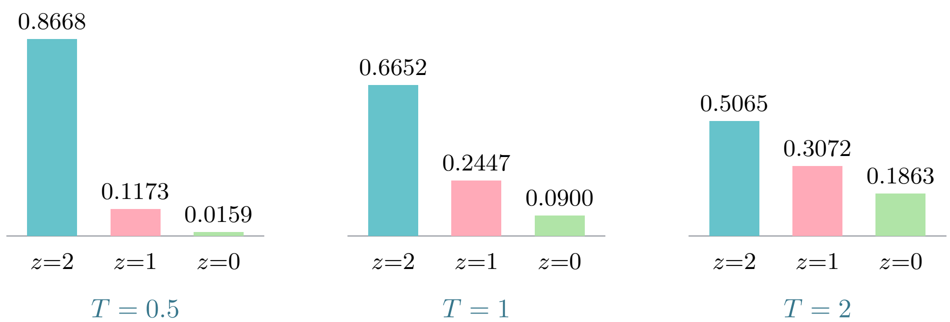
$$\mathrm{softmax}(z)_i \;=\; \frac{e^{z_i}}{\sum_j e^{z_j}}, \qquad \mathrm{softmax}(2, 1, 0) = (0.6652,\ 0.2447,\ 0.0900).$$

(The three printed values sum to 0.9999 — the exact shares sum to 1; four decimal places are a display, not the object.)

The exponential is not one choice among many. Demand of any smooth per-score recipe that a common shift $z + c$ leave the shares alone, and the exponential is forced: $e^{z_i + c} = e^c\, e^{z_i}$, and the common factor e^c cancels against the normalizer — exactly the same distribution. The demand pays immediately as a stability trick: since $\mathrm{softmax}(z) = \mathrm{softmax}(z - \max z)$, the overflowing scores $(1000, 1001, 1002)$ — naively ∞/∞, a row of NaNs — are rescued as $\mathrm{softmax}(-2, -1, 0)$, which is the workhorse's three shares in reverse order. Subtracting the max is a corollary of the invariance, not a hack. The name is Bridle's [33, 32]: a differentiable winner-take-all — "we like to refer to it as soft max".

10.5 Turning the dial

Softmax has a sharpness dial: divide the scores by a temperature T before the exponential.



On the workhorse scores, $T = 0.5$ sharpens the shares to $(0.8668,\ 0.1173,\ 0.0159)$ and $T = 2$ flattens them to $(0.5065,\ 0.3072,\ 0.1863)$ — and the winner never changes, at any $T > 0$, because exponentiation and positive scaling are monotone. The endpoints are limits, not settings: $T \to 0$ approaches one-hot (a clamped dial — the winner takes all), $T \to \infty$ approaches uniform.

The dial also answers a question the e-and-ln chapter left standing: why e? Run the same recipe in base 2 and the shares come out exactly $(4/7,\ 2/7,\ 1/7)$ — a *rational* softmax. But $b^z = e^{z \ln b}$: every base above 1 is base e at another temperature, $T = 1/\ln b$. Nothing forces e in the family; e is the unit in which the dial reads 1, because ln is the natural counter. And one caveat the dial itself proves: modern networks are overconfident, and fitting a single shared T recalibrates one *without changing any prediction* [21]. Numbers that can be rescaled wholesale were asserted, not measured.

10.6 The loss that trains

Join the arcs: score the machine by the log-likelihood it assigns the truth. With a single correct class c, the "cross-entropy loss" collapses to $-\ln \operatorname{softmax}(z)_c$ — negative log-likelihood, the alias acknowledged once. For softmax scores the loss is a visible gap on an owned ruler:

$$-\ln \mathrm{softmax}(z)_c \;=\; \mathrm{LSE}(z) - z_c,$$

the smooth max from the logarithms chapter minus the correct class's score [23]. On the workhorse, $\mathrm{LSE}(2, 1, 0) = 2.4076$, so the loss is 0.4076 if the truth scored 2, 1.4076 if it scored 1, 2.4076 if it scored 0 — one unit per rank the truth falls, the gap growing steadily as confidence points the wrong way.

Across frames, losses add — by the second section's logic, and under one license. Take a three-frame matrix of per-frame distributions over $\{A, B, \varepsilon\}$ (each column a softmax output, each summing to exactly 1):

$$\begin{pmatrix} 0.7 & 0.6 & 0.2 \\ 0.2 & 0.1 & 0.1 \\ 0.1 & 0.3 & 0.7 \end{pmatrix}$$

The path A, A, ε multiplies one entry per column: $0.7 \times 0.6 \times 0.7 = 0.294$ exactly — and the multiplication is legal only because the frames are *independent given the input*, the conditional chapter's exact lesson. Its logs add to -1.2242. This is one path's worth of the alignment chapter's sum: the trellis adds such products over every path that collapses to the transcript, and the CTC loss is that sum's negative log. The scale in the wild: Deep Speech runs this machine 29 ways per frame [8]; GPT-2 runs it 50,257 ways [1]. The per-frame scores the alignment chapter borrowed are now paid for: $y_t(k)$ is a softmax share, and the loss is a sum of gaps.

Where this goes. The next chapter differentiates everything this one built: the smooth max's sensitivities turn out to be the softmax shares themselves, and the gap loss's gradient is softmax output minus a one-hot target. Further down the road, CTC hands that same gradient a *soft* target — softmax output minus how often the truth used each cell — which is the identity the guide's closing chapters exist to reach.

Practice problems

Problem 10.1. Compute $\mathrm{softmax}(2, 1, 0)$ by hand to four decimal places ($e^2 \approx 7.3891$, $e^1 \approx 2.7183$). Why do your three printed values not sum to 1.0000 — and is that a defect?

Hint: Exponentiate, total, divide. Then think about what rounding three numbers separately does to their sum.

Problem 10.2. The naive normalizer $z/\sum_j z_j$ also produces numbers summing to 1. Evaluate it on $z = (2, 1, 0)$ and on the shifted $z' = (3, 2, 1)$, and name both failures. Then show in one line why softmax gives *exactly* the same distribution on z and z'.

Hint: A uniform shift adds no information — what should it do to the shares? For softmax, factor the shift out of every exponential.

Problem 10.3. Run the softmax recipe in base 2 on $z = (2, 1, 0)$: compute $2^{z_i}/\sum_j 2^{z_j}$ exactly. At what temperature T does ordinary (base-e) softmax reproduce your answer, and does the winner change at any base $b > 1$?

Hint: The powers of 2 are small integers. For the temperature, write $b^z = e^{z \ln b}$.

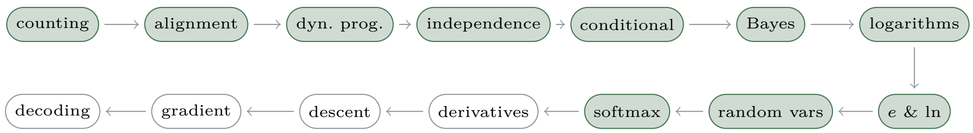
Problem 10.4. For the coin data $k = 3$ heads in $n = 4$ flips (count lens, $L(p) = 4p^3(1 - p)$): evaluate L at $p = 1/4$, $1/2$, and $3/4$; verify that the peak sits at $\hat{p} = k/n$; and compute the exact area under L over $[0, 1]$. What does that area prove?

Hint: The area is a polynomial integral — expand $4p^3(1 - p)$ first.

Problem 10.5. [stretch] On the three-frame matrix of the closing section, compute the exact probability of the path A, A, ε and the sum of its logs. Then the survival argument: how many frames of per-frame probability 0.1 does it take to kill a float32 product outright, and what does the sum of logs read at that point?

Hint: The product is three one-decimal factors. For the cliff, recall the logarithms chapter: float32's subnormal floor is $\approx 1.4 \times 10^{-45}$.

The road so far



The loss that trains is built: the smooth max's gap, summed over independent frames. Training means moving the scores to shrink it, and shrink-direction is a question about slope. The next chapter builds the toolkit — the slope as a function, the score trick that replaces heavy machinery, and the smooth max differentiated into shares you have already met.

Chapter 11

The derivative toolkit

The e-and-ln chapter built a device and never named its output: zoom in on a smooth curve until it is straight, read the slope, and watch the ratio settle as the step shrinks. This chapter names that ratio $\frac{d}{dx}$ and packs the smallest toolkit the road needs — the sum rule, the chain rule, two derivatives, and one reader called the score. The kit is deliberately tiny for a reason the logarithms chapter already gave: under the log, every product this road carries becomes a sum, so almost nothing else is required.

11.1 The slope is a function

Run the zoom at several points of one parabola $f(x) = x^2$ and plot the read-offs beneath it: every smooth curve carries a *second* curve, its slope at each point. At $x = 1$ the forward quotients $\big((1+h)^2 - 1\big)/h$ for $h = 1, \frac{1}{10}, \frac{1}{100}, \frac{1}{1000}$ read

$$3, \quad 2.1, \quad 2.01, \quad 2.001 \longrightarrow 2,$$

and no rounding is hiding anywhere: the quotient is exactly $2x + h$, so the entries are literally $2+h$, and "settling" means the h fading out of the answer. The dual graph keeps two numbers visibly different — at $x = 1$ the *height* is 1 while the *slope* is 2 — the confusion a derivative most often drowns in. One honest boundary: the device needs smooth. $|x|$ never straightens at 0, no matter the zoom. The notation is Leibniz's [29] — dy/dx in a manuscript of November 1675, in print by 1684 [59] — and this book commits to it because it makes the chain rule look like cancelling fractions, a promise examined two sections from now.

11.2 Nudge in, nudge out

For x^2 the derivative is drawn, not computed [16]. Draw the square of side x and grow the side by a nudge dx: the new area is the old square, plus two strips of size $x \cdot dx$, plus a corner of size $dx \cdot dx$. Halve the nudge and the strips halve — but the corner *quarters*: it is second-order small, and discarding it is honest bookkeeping, not a trick [51]. At $x = 3$ the strips say the slope is 6; the finite check $(3.01^2 - 9)/0.01 = 6.01$ agrees to the corner. The same fact wears a second costume: near $x = 1$ the map $x \mapsto x^2$ stretches the number line by $\times 2$, near $x = 3$ by $\times 6$ — a derivative is a *local stretch factor*, the seed the chain rule is about to harvest. Formula last:

$$d(x^2) \;=\; 2x\,dx + dx^2,$$

with the dx^2 dying for a visible reason.

11.3 Nudges add, nudges compose

Two rules carry the whole kit. *Changes add across a sum*: heights stack, so their changes stack — $d(f + g) = df + dg$, and constants ride along. And *rates multiply along a chain*: a nudge dx causes a nudge du, which causes a nudge dy, and the stretch factors compose,

$$\frac{dy}{dx} \;=\; \frac{dy}{du} \cdot \frac{du}{dx}.$$

Worked honestly on $y = (2x)^2$ at $x = 1$: the inner map doubles (rate 2), the outer map squares — at the inner *output* $u = 2$, rate $2u = 4$ — so the chain predicts $2 \times 4 = 8$. The forward quotients read 8.4, 8.04, 8.004 $\to$ 8. The table also refutes the classic mistake on contact: evaluating the outer rate at the input $x = 1$ instead of at $u = 2$ predicts 4, and 4 is not where the quotients are headed. The du cancellation is Leibniz's notation keeping its promise — but the number lines are the mechanism; the cancelling fraction is the mnemonic.

11.4 The curve that is its own slope

The e-and-ln chapter measured its mystery constants: 2^x grows at 0.6931 times its own height, 10^x at 2.3026 times, and the constants were the strides $\ln b$. In the new notation those measurements were derivatives all along: $\frac{d}{dx} b^x = \ln b \cdot b^x$, and e is the base whose constant is exactly 1 — by definition and by measurement, nothing proved twice. The ratio table $(e^{dt} - 1)/dt$ settles through 1.7183, 1.0517, 1.0050, 1.0005 $\to$ 1, so

$$\frac{d}{dx}e^x = e^x, \qquad \frac{d}{dx}\ln x = \frac{1}{x}.$$

The second is one chain-rule line through the undo trick: differentiate $e^{\ln x} = x$ to get $e^{\ln x} \cdot (\ln x)' = 1$, so $(\ln x)' = 1/x$ — the debt-repaid flip, differentiated. The mirror tells the same story in pictures: reflecting across $y = x$ swaps rise and run, so slopes reciprocate — at $x = e$ the slope of ln is $1/e$, mirroring e^x's slope e at height e. The attribution is quotable: Euler named e in 1748 (*Introductio* §122) [39] and wrote $d(e^x) = e^x\,dx$ in 1755 (*Institutiones* §188) [38].

11.5 Zero slope finds the peak

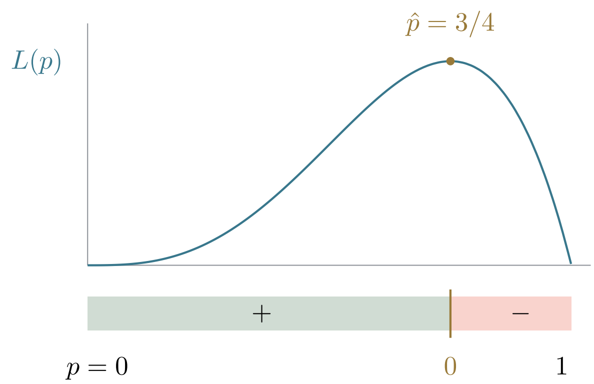
The softmax chapter found $\hat{p} = 3/4$ by sweeping a curve. The sweep is over. The tool is the *score function* — the counting strip differentiated: under ln, products become sums, so the derivative of a product becomes a sum of *relative rates*,

$$\frac{d}{dx}\ln f \;=\; \frac{f'}{f},$$

Euler's own sentence in §181 [38]. On the owned likelihood $L(p) = 4p^3(1-p)$: $\ln L = \ln 4 + 3\ln p + \ln(1-p)$ differentiates term by term — the sum rule, $\ln'$, and one chain-rule line for $\ln(1-p)$ — into the score

$$\frac{d\ln L}{dp} \;=\; \frac{3}{p} - \frac{1}{1-p}.$$

The score runs $+4$ at $p = 1/2$, $+0.9524$ at 0.7, 0 at 3/4, -1.25 at 0.8 — flipping sign exactly at the peak — and setting it to zero is a *linear* equation, $3(1-p) = p$, giving $\hat{p} = 3/4$. In general $k/p - (n-k)/(1-p) = 0$ solves to $\hat{p} = k/n$: the claim the random-variables chapter made about proportions is now a theorem.



Two honesty beats close the scene. The direct route $dL/dp = 4p^2(3-4p)$ finds the same zero at $3/4$ — plus a second root at $p = 0$ that is a valley floor, not a peak; and x^3's slope touches zero at the origin with no peak at all. Zero slope is necessary, never sufficient — the sign *change* is the certificate, and checking it is the habit that survives every tool built on top of this one. And the score itself needs $f > 0$: true for likelihoods away from the endpoints, where the log is defined.

11.6 The smooth max's shares

The closer differentiates the softmax chapter's own objects. Nudge $z_1 = 2$ by 0.01 and the smooth max $\mathrm{LSE}(2, 1, 0) = 2.4076$ moves by 0.00666 — which is 0.6652×0.01: e^2's share of the total. One chain rule and one sum rule prove the pattern general [36]:

$$\frac{\partial}{\partial z_i}\mathrm{LSE}(z) \;=\; \frac{e^{z_i}}{\sum_j e^{z_j}} \;=\; \mathrm{softmax}(z)_i.$$

The sensitivities of the smooth max *are* the softmax shares — the bars of the last chapter reborn as a gradient read-out, summing to 1 because the shares always did. And the loss follows in one line: with the correct class a scoring 2, $\mathrm{NLL} = \mathrm{LSE}(z) - z_a$ differentiates to

$$\nabla\,\mathrm{NLL} \;=\; (-0.3348,\ 0.2447,\ 0.0900) \;=\; p - \text{one-hot}:$$

the softmax shares with exactly 1 subtracted from the truth's entry. (Sign convention, spoken once: this is the gradient of the *loss* — descending it is ascending the likelihood.) As the correct score falls behind, the slope $p_c - 1$ walks -0.9100, -0.9868, -0.9993, pressing toward -1 — the softmax chapter's "roughly linear gap", now a theorem: a badly wrong confident answer buys loss at essentially one unit per unit of score.

Where this goes. A slope is advice: it says which way is downhill, and the next chapter builds the rule that takes the advice step by step. Beyond it, the road's destination hands this closing picture to every frame of CTC — the same p minus a target, but with the one-hot softened into how often the truth actually used each cell. The subtraction stays; only the target changes.

Practice problems

Problem 11.1. Write the forward quotient of $f(x) = x^2$ at $x = 1$ for a step h, simplify it exactly, and evaluate it at $h = 1, \frac{1}{10}, \frac{1}{100}$. What does the simplified form say "settling" means?

Hint: Expand $(1+h)^2$ before dividing.

Problem 11.2. Differentiate $y = (2x)^2$ at $x = 1$ by the chain rule, naming both rates and where each is evaluated. A classmate answers 4 — reconstruct their mistake, and refute it with the forward quotients at $h = \frac{1}{10}$ and $\frac{1}{100}$.

Hint: The outer rate is $2u$ — at which value of u?

Problem 11.3. Derive $\frac{d}{dx} \ln x = 1/x$ from $\frac{d}{dx} e^x = e^x$ using the undo identity $e^{\ln x} = x$, and check it numerically at $x = 2$ with the forward quotient at $h = \frac{1}{100}$.

Hint: Differentiate both sides of the identity; the chain rule does the rest.

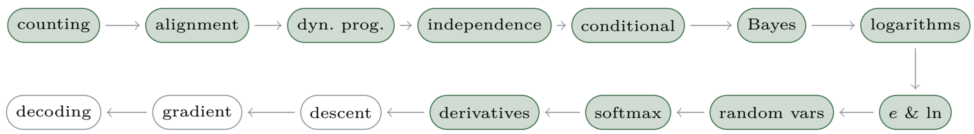
Problem 11.4. For the likelihood $L(p) = 4p^3(1-p)$: write $\ln L$, differentiate it term by term into the score, evaluate the score at $p = 1/2$, 0.7, and 0.8, and solve the zero. Then find the general $\hat{p}$ for k heads in n flips.

Hint: $\ln(1-p)$ needs one chain-rule line; the zero is a linear equation.

Problem 11.5. [stretch] With the correct class scoring 2 in $z = (2, 1, 0)$: differentiate $\mathrm{NLL}(z) = \mathrm{LSE}(z) - z_a$ component by component, and evaluate the gradient to four decimal places. Then let the correct class's score fall: on $z = (2, 1, t)$ with the truth in the third slot, compute the slope $\partial\,\mathrm{NLL}/\partial t$ at $t = 0, -2, -5$. What is the slope pressing toward, and why can it never arrive?

Hint: $\partial_i \mathrm{LSE} = \mathrm{softmax}(z)_i$, and z_a contributes -1 to exactly one component. For the walk, the slope is $p_c - 1$.

The road so far



The toolkit ended with a subtraction: softmax minus a one-hot target, the slope of the gap loss. A slope is advice — it says which way is downhill — and advice needs a rule for taking it. The next chapter is that rule: step, re-read the slope, step again; the learning rate that makes the walk converge, overshoot, or diverge; and the honest stopping story.

Chapter 12

Gradient descent

Count the road's inventory. A model's belief in a transcript is a number the trellis can price; the logarithm turned that belief into a loss worth minimising; softmax made the per-frame scores honest; and the derivative toolkit reads, at any setting of any knob, which way the loss tilts. One piece is missing between "here is a loss" and "the model trains": how a slope becomes an *update*. The answer is one move, almost embarrassingly small, applied over and over — and this chapter is about what the move does, what its single dial controls, and what its stopping place does and does not certify.

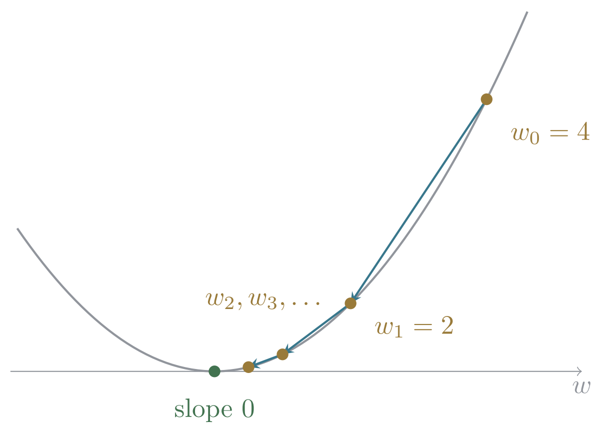
12.1 Walk downhill

Let a loss $L(w)$ depend on one knob w. The derivative chapter's central habit — the slope is itself a function $L'(w)$ — is the whole apparatus needed: at the current w, the sign of $L'(w)$ says which direction is uphill. So step the other way,

$$w \;\leftarrow\; w - \eta\, L'(w),$$

where η, the *learning rate*, converts a slope into a step length. That single line is gradient descent.

Run it exactly once by hand, on the plainest bowl there is: $L(w) = w^2$, so $L'(w) = 2w$. Take $\eta = \frac{1}{4}$ and start at $w_0 = 4$. The update is $w \leftarrow w - \frac{1}{2}w = \frac{w}{2}$: the walk is $4 \to 2 \to 1 \to \frac{1}{2} \to \cdots$, halving forever.



Watch what happened without being asked for. The direction was always downhill — the sign of the slope saw to that. And the steps shrank on their own: near the bottom the slope is small, so $\eta\, L'(w)$ is small. The rule brakes automatically, not because a schedule was imposed but because the landscape flattens where it matters. On this bowl, that is the entire story of convergence.

12.2 The learning rate is a bet

Keep the bowl, vary only η. The update is $w \leftarrow (1 - 2\eta)\, w$ — each step multiplies the distance to the bottom by a fixed factor:

$\eta = \frac{1}{4}$	factor $\frac{1}{2}$	glides in
$\eta = \frac{3}{4}$	factor $-\frac{1}{2}$	overshoots the bottom each step, yet converges
$\eta = 1$	factor -1	ping-pongs between 4 and -4 forever
$\eta = \frac{5}{4}$	factor $-\frac{3}{2}$	every "descent" step lands higher: diverges

The dial has a cliff, not a dimmer. Too small merely wastes steps; too large converts descent into ascent, and nothing inside the rule warns you — the update only ever reads the slope *where it stands*, while the failure is a property of how sharply the landscape curves across the whole step. The learning rate is a bet about curvature, and the bowl collects on bad bets immediately.

12.3 Where the walk stops

The update is zero exactly where $L'(w) = 0$: gradient descent stops at flat ground, full stop. Whether flat ground is the *bottom* is a separate question, and the derivative chapter already issued the right habit: look at the sign change. Slope crossing $-$ to $+$ is a valley; $+$ to $-$ is a hilltop; no crossing is a shelf. Descent stalls at every one of them. On the double well $L(w) = \frac{w^4}{4} - \frac{w^2}{2}$, whose slope $w^3 - w$ vanishes at -1, 0, 1, a walk started anywhere to the right of 0 settles into the valley at 1 — but a walk started *exactly* at 0 never moves: it sits on the hilltop, gradient zero, perfectly stalled, certifying nothing. A zero gradient is where the walk ends; the sign change around it is what tells you what was found. (The knife-edge is fragile — nudge w_0 to 0.1 and the walk falls into a valley — which is exactly why the failure is instructive rather than common.)

12.4 Many knobs, and the road's own loss

A real model has many knobs at once, and the generalisation costs one sentence: the *gradient* is the vector collecting the loss's slope along every knob, and the update steps against it, $\mathbf{w} \leftarrow \mathbf{w} - \eta \nabla L(\mathbf{w})$. Each knob reads its own partial slope; all move together. That — nothing more — is *plain gradient descent*, and it is the engine under essentially all of deep learning [23].

The road can afford to watch it run on its own worked example. Make the alignment chapter's four-frame score table trainable — each frame's three scores driven through softmax from free knobs — and let the loss be the negative log of the alignment sum, $-\ln P(\mathrm{AB} \mid X)$, which starts at 0.7181. Plain gradient descent at $\eta = 1$ walks it down to 0.1602 after ten steps, 0.0356 after fifty, 0.0088 after two hundred, and 0.0003 after five thousand. Read the walk's shape, not just its endpoint: over three quarters of the loss vanished in the first ten steps, and the last leg spent nearly five thousand steps buying less than a hundredth. That is the parabola's lesson at scale — flattening loss means shrinking gradient means shrinking steps — and it is why a training curve's long, nearly level tail is normal physiology, not a stall.

One honesty note, matching the video road's own scope: everything real training adds — minibatch stochasticity, momentum, adaptive per-knob steps, schedules — is refinement layered on this one rule, and none of it appears on this road. What runs here, and in every training curve this book shows, is the bare update, undoped. Plain gradient descent is enough to train the road's worked example to convergence; the refinements exist because "many knobs" in practice means billions, not twelve.

Where this goes. This chapter treated one object as given: the gradient of the alignment sum itself — a loss that is a sum over 15 paths needs its slope organised as cleverly as its value was. The next chapter computes exactly that, and the training beats of its parent series ("plain gradient descent", named on screen) run precisely the walk quoted above. Farther out, descent has earned its own future video series — learning-rate cliffs, valleys and hilltops, the walk watched live — and this chapter is that series' seed.

Practice problems

Problem 12.1. On $L(w) = w^2$ with $\eta = \frac{1}{4}$ and $w_0 = 4$, write out w_1, w_2, w_3 exactly. After how many steps does $|w_k|$ first drop below 0.01?

Hint: Substitute $L'(w) = 2w$ into the update and simplify before iterating.

Problem 12.2. Still on $L(w) = w^2$ from $w_0 = 4$: for $\eta \in \{\frac{1}{4}, \frac{3}{4}, 1, \frac{5}{4}\}$, find the per-step factor and classify each walk (converges, oscillates and converges, oscillates forever, diverges). For which η does the walk reach the bottom at all?

Hint: Write the update as $w \leftarrow (1 - 2\eta)w$ and ask when the factor's magnitude is below 1.

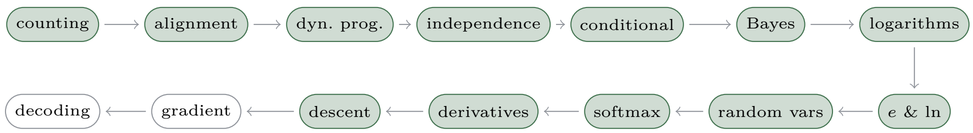
Problem 12.3. On the double well $L(w) = \frac{w^4}{4} - \frac{w^2}{2}$ with $\eta = 0.1$: where does the walk settle from $w_0 = 0.5$, from $w_0 = -0.5$, and from $w_0 = 0$? Classify each stopping place with the sign-change test.

Hint: $L'(w) = w^3 - w$ vanishes at three points; check the slope's sign just left and right of each.

Problem 12.4. [stretch] From the walk in the chapter — loss 0.7181 at the start, 0.1602 after 10 steps, 0.0088 after 200, 0.0003 after 5000 — compute the average per-step shrink factor of the loss over the first ten steps, and over the last 4800. What do the two factors say about where a training run spends its time?

Hint: If ten steps multiply the loss by r^{10}, the average factor is the tenth root of the ratio.

The road so far



The walker knows how to step. What it needs now is the slope of THE loss — the alignment sum over
anchor001.paths15 paths — and a sum too wide to list needs its slope organised as cleverly as its value was. The next chapter is the destination: the trellis swept backward, every cell priced by the paths through it, and the gradient collapsing to one subtraction — softmax minus how often the truth used each cell.

Chapter 13

The CTC gradient: softmax minus occupancy

This is what the road was for. The alignment chapter left a loss — the negative log of a sum over every path that spells the transcript — and every chapter since has been quietly forging a tool to differentiate it. The worked example returns one last time: transcript $Y = \mathrm{AB}$, alphabet $\{A, B, \varepsilon\}$, $T = 4$ frames, the wrapped states $(\varepsilon, A, \varepsilon, B, \varepsilon)$, the 15 paths — and now a concrete per-frame matrix (rows $A/B/\varepsilon$, one column per frame): $0.7/0.2/0.1 \cdot 0.6/0.1/0.3 \cdot 0.2/0.1/0.7 \cdot 0.1/0.7/0.2$. On it, exact arithmetic gives $P(\mathrm{AB} \mid X) = 0.4877$ and loss $-\ln P = 0.7181$. By the chapter's end, the gradient of that loss at every one of the twelve score cells will be a subtraction you can read off a picture.

13.1 The other half of the trellis

The forward variable $\alpha_t(s)$ answered "how did we get here?". Its mirror $\beta_t(s)$ answers "how do we finish?" — the weight of every legal way to complete the transcript from state s after frame t. Same grid, same edges, swept against the arrows.

One bookkeeping decision matters more than it looks: each frame's emission is a coin exactly one of the two variables may pocket. This book adopts the convention of Graves's 2012 treatment [7]: α's column pockets frame t's emission, and β starts its product at $t+1$ — so at the last frame, $\beta_T = 1$ at the two accepting states and 0 elsewhere, and no emission is ever counted twice. At unit weights the backward sweep is pure counting: its first column reads $(5, 10, 6, 4, 1)$ down the five states, and the two *initial* states give

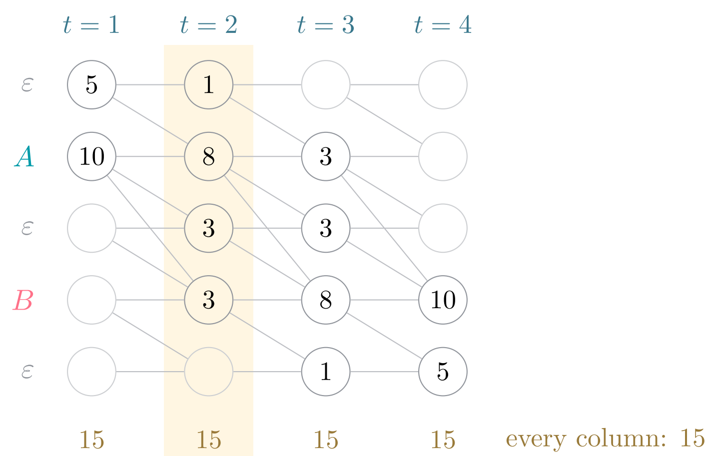
$5+10=15$ — the flagship count, found now from the far end. (On this example the backward columns mirror the forward ones — a symmetry of this palindromic transcript, not a theorem.) Formula last, the mirror recurrence:

$$\beta_t(s) \;=\; \sum_i \beta_{t+1}(s{+}i)\; y_{t+1}(z'_{s+i}),$$

the sum over stay, advance, and the skip where the grid allows it. This is the backward half of forward–backward — the HMM lineage every CTC implementation inherits [37].

13.2 Paths through a cell

Multiply the two halves at any cell and something new appears: $\alpha_t(s)\,\beta_t(s)$ is the total weight of every path that passes *through* that cell — ways in, times ways out, the counting grid's product rule reborn carrying probability. At unit weights, cell $(t{=}2, A)$ has 2 prefixes and 4 suffixes: $2 \times 4 = 8$ of the 15 paths cross it.



Sweep a column across the grid and the series' strongest device appears — Eisner's constant column [28]: at unit weights *every* column of products sums to 15, and on the real matrix every column sums to the same $P(\mathrm{AB} \mid X) = 0.4877$ — four columns, one answer, which also equals α's two-final-state sum and β's two-initial-state sum. The reason is a sentence: every path crosses every column exactly once. Formula last:

$$P(Y \mid X) \;=\; \sum_s \alpha_t(s)\,\beta_t(s) \qquad \text{for any } t.$$

The column is also the bookkeeping self-check: pocket an emission twice and its column's sum scales visibly out of line.

13.3 Where the truth spends its time

Divide each column by its own sum and it becomes a distribution:

$$\gamma_t(s) \;=\; \frac{\alpha_t(s)\,\beta_t(s)}{P(Y \mid X)},$$

the share of the truth's probability passing through each cell — *where the truth spends frame* t. The divide-by-P is the conditional chapter's renormalized slice, conditioned this time on the transcript; columns sum to 1 because each path occupies exactly one cell per frame — the topology grants it. Folding blank's three grid rows into one class (a summation drawn exactly once — the grid has five rows, the bars three) gives the per-class occupancy on the real matrix:

	$t=1$	$t=2$	$t=3$	$t=4$
A	0.9028	0.7086	0.1464	0
B	0	0.0330	0.1403	0.9487
ε	0.0972	0.2584	0.7133	0.0513

Columns are distributions; *rows are not.* A's row sums to 1.7578 — an expected *dwell time*: of the four frames, the truth spends nearly two of them emitting A, and the three dwells $1.7578 + 1.1220 + 1.1202$ sum to exactly $4 = T$. The balance point from the random-variables chapter has returned: occupancy is an expectation over the posterior path distribution. One degenerate setting shows what the object is made of: under *uniform* outputs ($y = 1/3$ everywhere), $P = 5/27$ and γ collapses to pure path counts over 15 — the alignment chapter's counting scene reborn as a distribution. Stand a γ column beside the derivative chapter's one-hot target and the chapter's destination is already visible: this is a target *gone soft*.

13.4 The sensitivity of the sum

Now differentiate the monster — and find there is no monster. First, a linearity observation: each path uses frame t exactly once, so any single cell's score $y_t(k)$ appears in each path's product to power 0 or 1 — P is *linear* in that one cell, and its slope is the cell's coefficient, which the waist picture already named: $\alpha\beta$ summed over the cell's states.

The load-bearing step is the log-sensitivity reading, and it is the derivative chapter's closing move at path scale. Nudge one cell multiplicatively — scale $y_t(k)$ by $(1+h)$ — and every path through the cell scales by the same factor while every other path stands still. So the *relative* change of P is the share of P passing through the cell:

$$\frac{\partial \ln P}{\partial \ln y_t(k)} \;=\; \gamma_t(k),$$

with the score function $d \ln f = f'/f$ doing the bookkeeping, as it has since the toolkit chapter. In LSE's gradient each term owned its own score; here many paths share one cell, so their shares *add* into occupancy — the sensitivity of a smooth max is a share, and the sensitivity of a path-sum is a pooled share. Since the loss is $L = -\ln P$, the formula lands last:

$$\frac{\partial L}{\partial \ln y_t(k)} \;=\; -\gamma_t(k).$$

Nudge one cell's log-probability, and the loss moves by exactly that cell's occupancy.

13.5 Softmax minus occupancy

The per-frame y is a softmax, so one owned step remains. The log-softmax derivative — one line from $\partial_i \mathrm{LSE} = \mathrm{softmax}_i$ — is $\partial \ln y_j / \partial u_k = \delta_{jk} - y_k$, where u are the frame's logits. Chain it against $-\gamma$:

$$\frac{\partial L}{\partial u_t(k)} \;=\; -\sum_j \gamma_t(j)\left(\delta_{jk} - y_t(k)\right) \;=\; -\gamma_t(k) + y_t(k)\sum_j \gamma_t(j),$$

and the checksum $\sum_j \gamma_t(j) = 1$ — the constant column, paying its second time, now load-bearing in the algebra — collapses the whole thing:

$$\boxed{\frac{\partial L}{\partial u_t(k)} \;=\; y_t(k) - \gamma_t(k)}$$

(Graves 2012, eq. 7.34 [7]). The per-frame gradient of the CTC loss is the softmax output minus the truth's occupancy. On the worked matrix:

$y-\gamma$	$t=1$	$t=2$	$t=3$	$t=4$
A	-0.2028	-0.1086	$+0.0536$	$+0.1000$
B	$+0.2000$	$+0.0670$	-0.0403	-0.2487
ε	$+0.0028$	$+0.0416$	-0.0133	$+0.1487$

Every column sums to 0 — mass pushed off the truth's cells lands somewhere — and on this example even the printed four-decimal digits sum to 0.0000 exactly. Two cells are worth reading. At $t = 1$, B's entry is *exactly* $+0.2000 = y$: the state is unreachable, its occupancy is 0, and the gradient is the output itself — pure push-down. And scoring the *wrong* transcript BA against the same matrix flips t=1, A to exactly +0.7000 for the same reason: an alignment the transcript cannot use gets no protection.

The degeneration closes the derivative chapter's promise: let one path carry everything and γ's columns snap one-hot — $y - \gamma$ becomes $p -$ one-hot, the $(-0.3348,\ 0.2447,\ 0.0900)$ of the toolkit's closer, received as the special case. Bridle's "output minus a one-from-N target" [33], with the one-from-N gone soft. And one implementation trap, because the identity is what CTC backward passes hard-code: the clean form lives at the *logits*. Hand the loss anything but a true log-softmax and the forward loss stays right while the gradient goes silently wrong [48]. (The y-space form $-\gamma/y$ is all-negative — it only pushes up — which is why the u-form is the teachable one.)

13.6 The error signal learns

Watch the identity during training. Graves's original figure traced the error signal through three stages [17]; here the arc is rebuilt countable, by plain gradient descent — the previous chapter's walker, learning rate 1.0 — on free per-frame logits. (One sign note where plots enter: error panels traditionally plot the *push* $\gamma - y$, the negative of the gradient — above the axis means "raise this output". Columns balance to zero in either sign.)

Diffuse. Start uninformative: uniform outputs. Then γ is exactly path counts over 15 — the input never consulted — and the gradient is pure fractions: at $t = 1$, $y - \gamma = (-1/3,\ +1/3,\ 0)$ across (A, B, ε). The error is determined by the target sequence alone, spread wide and time-symmetric.

Localised. Descend from the worked matrix: the loss walks 0.7181 $\to$ 0.1602 after ten steps $\to$ 0.0356 after fifty. The error concentrates where the outputs already lean — the pushes sharpen around the emerging alignment A, A, ε, B.

Gone. By five thousand steps the loss reads 0.0003 and the bars sink into the axis. The teaching gem hides at frame 3, which converges not to a spike but to the *mixed* output $(0.032,\ 0.218,\ 0.750)$ with gradient $\to 0$: once frames $1, 2, 4$ say A, A, B, all three of frame 3's choices collapse to AB, so the loss goes indifferent — y matches γ out of indifference, not certainty. A vanished gradient does not mean one-hot outputs; it means the outputs agree with the occupancy, which is a weaker and more interesting thing.

13.7 Why the spikes appear

Trained CTC models emit sharp single-frame spikes separated by long runs of blank, and the alignment chapter warned: never read the spikes as segment boundaries. Here, finally, is the mechanism — and it is topology plus weight sharing, not acoustics.

First, topology. Count label occurrences over all AB-paths, with the input nowhere in the computation. At $T = 4$ the counts across (A, B, ε) are $(21,\ 21,\ 18)$ — blank is *not* dominant; the boundary case is honest. At $T = 5$ (now 35 paths — the count the alignment chapter's problem set met) they are $(56,\ 56,\ 63)$, and blank is ahead for good: its share grows with T, toward $2/3$ in the one-letter limit under uniform outputs. Under uniform initialization the occupancy *is* these counts — so the earliest training signal already tilts blankward for every T past the boundary.

Second, weight sharing. Free per-frame outputs never spike — the worked example converged to an honest alignment above. But force one shared softmax to serve every frame (a bias-only model) and descent tells a different story: at $T = 4$ it settles at the honest $(0.4,\ 0.4,\ 0.2)$; at $T = 12$ it converges to $(0.0919,\ 0.0919,\ 0.8162)$ — argmax blank at every frame, an *empty* decode, 100% error — and this is a *local* optimum: the global optimum has zero error. That trap is proved, for a feed-forward network from uniform initialization, by Zeyer, Schlüter and Ney (arXiv preprint) [63], who also name the repair: a label prior folded into the loss — peakiness is a steerable training artifact, never a timestamp.

The closing map: the identity is a family. One-hot cross-entropy (γ one-hot — the degenerate case), distillation (a teacher's soft targets), CTC (γ the occupancy) — all one gradient shape, output minus target, with only the target changing. And γ itself is the soft alignment real systems compute — the E-step object of the HMM tradition, met here as the thing the gradient reads off the trellis.

Where this goes. Training is solved; reading the trained model's output stream is not. Greedy decoding has been suspect since the alignment chapter, spikes are not segmentations, and the honest decoder — beam search over *collapsed prefixes*, each prefix carrying two probabilities — is the road's last construction: the next chapter closes the loop from training to deployment.

Practice problems

Problem 13.1. At unit weights, run the backward recurrence on the AB trellis ($T = 4$, states $\varepsilon, A, \varepsilon, B, \varepsilon$): $\beta_4 = 1$ at the two accepting states, 0 elsewhere. Compute the $t = 1$ column of suffix counts, and check its two initial states against the flagship path count.

Hint: Work right to left; a state's count is the sum of its successors' counts (stay, advance, and the skip only over a blank between different letters).

Problem 13.2. Using the forward counts $(1, 1, 0, 0, 0)$, $(1, 2, 1, 1, 0)$, $(1, 3, 3, 4, 1)$, $(1, 4, 6, 10, 5)$ and your backward counts, form the per-cell products $\alpha\beta$ at every cell and sum each column. What do the four sums say, and why must they agree?

Hint: Backward columns: t=1 $(5, 10, 6, 4, 1)$, t=2 $(1, 4, 3, 3, 1)$, t=3 $(0, 1, 1, 2, 1)$, t=4 $(0, 0, 0, 1, 1)$.

Problem 13.3. Under uniform outputs ($y_t(k) = 1/3$ for all t, k): compute $P(\text{AB} \mid X)$, the occupancy column at $t = 1$, and the gradient $y - \gamma$ there. What is remarkable about what the computation never used?

Hint: Every path has the same weight, so γ is a ratio of counts.

Problem 13.4. On the worked matrix, frame 1 has $y_1 = (0.7, 0.2, 0.1)$ and occupancy $\gamma_1 = (0.9028, 0, 0.0972)$. Write the gradient column, verify it sums to zero, and explain why B's entry equals its softmax output exactly — no rounding involved.

Hint: Which trellis states can a path for AB occupy at frame 1?

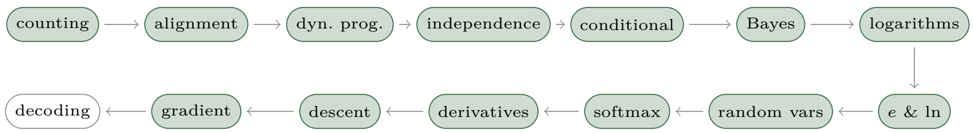
Problem 13.5. Derive the identity. Starting from $\partial L/\partial \ln y_t(k) = -\gamma_t(k)$ and the log-softmax derivative $\partial \ln y_j/\partial u_k = \delta_{jk} - y_k$, show $\partial L/\partial u_t(k) = y_t(k) - \gamma_t(k)$, naming the step where the constant column does the work.

Hint: Chain over j, then split the sum into its δ term and its y_k term.

Problem 13.6. [stretch] Count label cells over all AB-paths — with no input anywhere — at $T = 4$ and $T = 5$: how many of the path-cells carry A, B, and blank? At which T does blank take the lead, and what does that say about the first gradient step from an uninformative start?

Hint: 15 paths of 4 cells, then 35 paths of 5 cells; enumerate or use the unit-weight γ columns, which are exactly these counts over the path total.

The road so far



Training is solved: the error signal breathes, localises, dies. But a deployed model is read, not trained — and reading the output stream honestly is its own problem, suspect since the alignment chapter proved greedy wrong. The last chapter closes the loop: from best-path to collapsed prefixes, two ledgers per prefix, and the spikes read for what they are.

Chapter 14

Decoding: from scores to a transcript

Every chapter so far has served training: score a transcript, take the log, follow the gradient downhill. But nobody deploys a loss. A model ships, a clip arrives, and the user wants words — no reference transcript in sight. That inverse problem is *decoding*: given the per-frame scores y_t, return the transcript the model believes, ideally $\operatorname{argmax}_Y P(Y \mid X)$ with P the alignment chapter's sum over paths. Training never had to search — the transcript was handed to it. Decoding is nothing but search, and this chapter closes the road's loop — train, decode, deploy — with the two searches that matter: the cheap one, and the honest one.

14.1 Best path: decode the way the frames vote

The obvious decoder takes each frame's favourite symbol and collapses the result through the map $\mathcal{B}$ from the alignment chapter — merge repeats, drop blanks. This is *best-path* (greedy) decoding [17]: T table lookups and a collapse, fast enough for anything.

What it finds, though, is the most probable *path* — and the road learned in the alignment chapter that transcripts and paths part company. The construction that justified training on the sum returns here, wearing its deployment costume. Two frames, $y_t(A) = 0.4$ and $y_t(\varepsilon) = 0.6$ at both: greedy takes the blank twice and announces the *empty* transcript, whose single path carries mass 0.36. The transcript A pools its three paths to 0.64 — nearly two to one against greedy's answer. The model, read correctly, heard an A; read greedily, it heard nothing. The mismatch is sum-versus-max: greedy crowns the heaviest single path, while the truth belongs to the transcript whose *team* of paths is heaviest.

When does greedy get away with it? When one path hoards the mass — and the gradient chapter showed that trained CTC models end up exactly there: peaked, blank-heavy outputs where the argmax path towers over the field. That is why greedy is the production default and usually right. But the hedging inputs — accents, noise, the audio that actually needed a good model — are precisely where the mass spreads across paths and greedy starts lying.

14.2 Searching transcripts: the collapsed-prefix beam

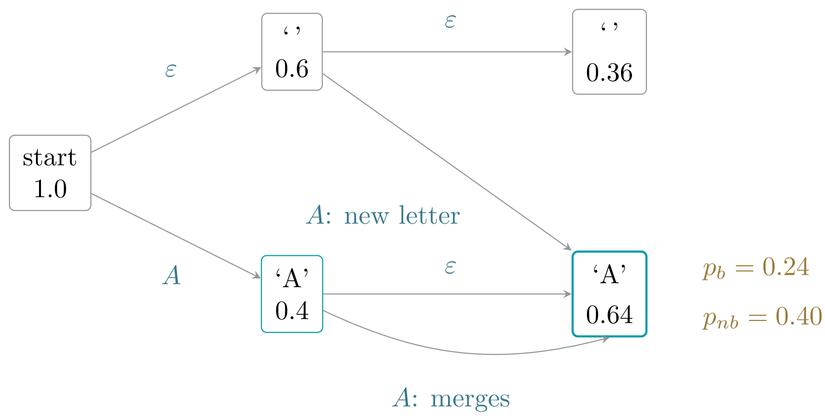
Doing better means searching over transcripts, and the transcript space is exponential — enumeration died in the alignment chapter and stays dead. The standard compromise is a *beam*: march through the frames keeping only the k most massive candidates, extending each by one frame at a time. For CTC the candidates must be *transcript prefixes*, not paths — and that single choice brings the whole road along with it. Many paths spell the same prefix, so as the beam extends, path masses pouring into the same prefix are *merged* — the dynamic-programming move again, shared prefixes stored once, now running on a pruned tree of candidates [22].

One subtlety makes CTC's beam different from every textbook beam, and it is the collapse map's fault. A prefix must carry *two* masses, not one:

$$
\begin{aligned}
p_b(\text{prefix}) &= \text{the mass of its paths ending in blank,} \\
p_{nb}(\text{prefix}) &= \text{the mass of those ending in its last letter,}
\end{aligned}
$$

because the two halves have different futures. Append the letter c to a path that already ends in c: the repeat merges and the transcript does not grow — *unless* a blank intervened, in which case a genuine new c begins. Only the p_b share can open a double letter; only the p_{nb} share merges silently. Collapse the two numbers into one and the beam can no longer tell which extensions are legal — it must overcount or undercount, and the problems below make it do both.

Run the beam on the construction above, width 2 (which here keeps every candidate, so the answer is exact):



Three streams meet in the prefix A at the second frame: a new letter opened from the empty prefix, a blank extension of A (which parks its mass in p_b, holding the door open for a future double letter), and the silent merge of A's repeat. The totals — 0.36 against $0.24 + 0.40 = 0.64$ — are exactly the alignment chapter's sums: the beam with its two ledgers is the forward recurrence wearing a search harness, and pruning is the only approximation it makes.

14.3 The spikes are still not timestamps

One inherited caution, restated at the door of deployment. The gradient chapter explained why a trained model's letters fire as one-frame spikes fenced by blank [63]: the loss summed over every alignment and never paid for timing. The decoding consequence: the frame at which the decoder's A fires is not where the A was said — not its onset, not its centre, nothing calibrated at all. A decoder returns *what* was said; if a product needs *when*, that is forced alignment, a different tool answering a question this loss never asked.

14.4 The loop, closed

Train on the sum; decode greedily when the outputs are peaked, with a beam when hedges matter; deploy. The beam brings one more gift that production systems lean on daily: its per-extension scoring is a natural place to splice in an external language model, patching exactly the hole the alignment chapter declared — frames independent given the input, language left on the table [8]. A transcript then wins by sounding right to the acoustic model *and* reading right to the language model, and the blend is tuned, not derived.

Where this goes. Nowhere — and that is the point: this is the road's last chapter, and the loop from waveform to weights to words is closed. What is deliberately left folded is the beam itself: widths and pruning, the log-space ledgers, language-model fusion and its tuning knobs belong to a future beam-search video series, and this chapter's sketch is that series' seed.

Practice problems

Problem 14.1. Best-path decode this three-frame model over $\{A, B, \varepsilon\}$: $y_1 = (0.5, 0.2, 0.3)$, $y_2 = (0.4, 0.3, 0.3)$, $y_3 = (0.2, 0.3, 0.5)$ (listed as A, B, ε). Then decide: does the full sum over paths crown the same transcript here?
Hint: Argmax each frame, collapse; for the second half, no enumeration is needed by hand — ask which transcripts could plausibly compete, or trust the committed script's brute force.

Problem 14.2. Generalise the chapter's construction: two frames, $y_t(\varepsilon) = q$, $y_t(A) = 1 - q$. For which q does greedy decode the empty transcript while the sum prefers A? Locate the exact boundary.
Hint: Greedy follows the frame favourite; the two transcript masses are q^2 and $1 - q^2$.

Problem 14.3. Run the two-ledger beam by hand on the chapter's construction ($y_t(A) = 0.4$, $y_t(\varepsilon) = 0.6$, $T = 2$, width 2), tracking (p_b, p_{nb}) for every prefix at each frame. Confirm the final masses against the alignment chapter's sums.
Hint: Frame one leaves two prefixes. At frame two, the prefix A receives three contributions — route each into the correct ledger.

Problem 14.4. [stretch] Uniform coin frames: $y_t(A) = y_t(\varepsilon) = 0.5$ over $T = 3$. Compute the true $P(\text{AA})$, then show what a one-ledger beam — a single mass per prefix, so every letter extension is treated as a new letter — would report for AA, and diagnose the error.
Hint: Which length-3 paths collapse to AA? The double letter needs its separating blank.

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